VIETNAM NATIONAL UNIVERSITY – HO CHI MINH CITY
UNIVERSITY OF SCIENCE
FACULTY OF INFORMATION TECHNOLOGY

TECHNICAL REPORT

Lab 2: Decision Tree Modeling and Improvement
Predicting Student Dropout and Academic Success

Course: Introduction to Artificial Intelligence

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Group 2

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a. Group Introduction

Group 2

Table 1: Team members and contributions

Name	Student ID	Contribution	%
Trần Hoàng Phúc	24127505	Data collection & description, EDA, data preprocessing pipeline, final results aggregation, wrote Introduction / Dataset Description / Comparison / Conclusion	20%
Nguyễn Văn Minh	24127205	Baseline model, tree/rule visualization tooling, wrote Baseline Model & Analysis of the Tree	20%
Thái Quang Huy	24127177	Cost-complexity pruning (Improvement 1), wrote section f.1	20%
Nguyễn Hữu Gia Minh	24127078	Class imbalance handling (Improvement 2), wrote section f.2	20%
Mai Phương Thùy	24127249	Feature selection for early warning (Improvement 3), slides, video, references, wrote section f.3	20%

Each member completed one defined project workstream. The five equal shares therefore allocate 20% to each member and sum to 100%.

b. Introduction

b.1 Brief introduction to decision trees

A decision tree is a supervised learning model used for both classification and regression tasks. It represents the decision process as a tree structure: each *internal node* tests one feature (e.g. “Tuition fees up to date ≤ 0.5?”), each *branch* corresponds to the outcome of that test, and each *leaf node* assigns a predicted class.

Training a decision tree follows a **greedy, top-down, recursive-partitioning** strategy: at each node, the algorithm evaluates every feature and every possible split threshold, and picks the split that reduces class *impurity* the most, then repeats recursively on each resulting subset. Two impurity measures are standard:

- **Entropy / Information Gain**: used by the ID3 algorithm (Quinlan, 1986), measuring the information gained by a split.
- **Gini impurity**: used by CART (Breiman et al., 1984), measuring the probability of misclassification if a label were drawn at random from the node’s class distribution.

The scikit-learn library used in this project implements an optimized variant of CART (Pedregosa et al., 2011); it defaults to Gini but supports switching to entropy via the `criterion` parameter.

The **main strength** of a decision tree compared to “black-box” models (neural networks, complex ensembles) is **interpretability**: every path from the root to a leaf can be read directly as an `IF (condition 1) AND (condition 2) ... THEN (label)` rule, with no separate explanation technique (SHAP, LIME) required. Once nominal categories have been encoded, trees also handle the resulting numerical and indicator features without feature scaling. Both properties are used directly in Section c below.

The **main weakness** is a tendency to **overfit**: an unconstrained tree keeps splitting until almost every leaf is pure on the training set, effectively memorizing training noise, which yields very high training accuracy but a much lower test accuracy (Mitchell, 1997; Breiman et al., 1984). Section d shows this pattern directly in the baseline model, which motivates the pruning improvement in Section f.

b.2 Objective of the project

The project applies this theory to a real, socially meaningful problem: **predicting whether a student drops out, remains enrolled longer than expected, or graduates on time**, using real data from 4,424 university students in Portugal. Concretely, the group:

1. Builds and evaluates an unconstrained baseline decision tree, measuring performance with the full set of metrics appropriate for a 3-class classification problem (Accuracy, Precision, Recall, F1-score, ROC-AUC, Confusion Matrix).

2. Analyzes the baseline tree's structure: what the root split is and why the algorithm chose it, whether the tree shows signs of overfitting, and extracts concrete decision rules.

3. Proposes and tests three improvement methods, each targeting a different weakness of the baseline:

   - **Pruning** (cost-complexity pruning): trades tree complexity for generalization to address overfitting.
   - **Class-imbalance handling** (`class_weight='balanced'`, SMOTE): addresses the model's default bias toward the majority class (Graduate). At baseline, Enrolled recall (0.3836) is well below Dropout's already-reasonable recall (0.6796), so Enrolled is the main target of this improvement.
   - **Feature selection for early warning**: addresses a practical limitation. Semester outcomes are unavailable at enrollment, so the model with the highest accuracy is not necessarily the model best suited to that earliest intervention point. Operational feasibility of the retained variables remains conditional on the feature-availability assumption stated in Section f.3.

4. Quantitatively compares all five model configurations and asks **which model suits which application goal**, in line with the assignment brief's emphasis on explaining how the tree is built, how it behaves, and why each improvement helps or does not, rather than only reporting a final accuracy score.

c. Dataset Description

c.1 Data source

The group uses the **"Predict Students' Dropout and Academic Success"** dataset from the UCI Machine Learning Repository (ID = 697), published by Realinho, Vieira Martins, Machado, and Baptista (Realinho et al., 2021), licensed CC BY 4.0. The dataset is described in detail in the accompanying paper (Realinho et al., 2022).

The data was collected at **Instituto Politécnico de Portalegre**, Portugal, spanning **17 undergraduate degree programs** (Agronomy, Design, Nursing, Journalism and Communication, Management, Social Service, Biofuel Production Technologies, Informatics Engineering, etc.), academic years **2008/09 to 2018/19**. It combines information from several administrative sources at enrollment time (admission records, socio-economic data) with academic outcomes at the end of each of the first two semesters (Realinho et al., 2022, Section 2, "Data Description").

c.2 Dataset description — numbers confirmed from actual data

The group did not copy figures from the UCI metadata page; instead it directly inspected the raw dataset to confirm:

- The raw file contains **4,424 rows and 37 columns**: 4,424 samples (students), 36 features plus 1 label column `Target`.
- There are **zero** missing values across the whole dataset (the authors pre-cleaned the data).
- **Notable finding:** the UCI page lists 36 features, but Table 1 of the original paper lists only 34. Cross-checking the actual column list against that table, the discrepancy is exactly two columns: `Previous qualification (grade)` and `Admission grade`, both present in the released CSV but absent from the paper's table. That discrepancy is only visible by reading the actual column list, not by copying the metadata page.

The 36 features fall into six groups by nature: demographics (6 columns, e.g. Gender, Age at enrollment), socio-economic status (8 columns, e.g. Tuition fees up to date, Scholarship holder), macroeconomics (3 columns: Unemployment rate, Inflation rate, GDP), academic background at enrollment (7 columns, e.g. Course, Admission grade), first-semester outcomes (6 columns), and second-semester outcomes (6 columns).

Target variable has 3 classes, moderately imbalanced:

Table 2: Target class distribution (n = 4,424)

Class	Count	%	Meaning
Graduate	2,209	49.9%	Graduated on the standard timeline (3 years; 4 for Nursing)
Dropout	1,421	32.1%	Dropped out
Enrolled	794	17.9%	Still enrolled past the standard timeline

Percentages are rounded to one decimal place and therefore sum to 99.9% rather than 100%; the unrounded values (49.93%, 32.12%, 17.95%) sum to 100%.

Enrolled is both the minority class and the hardest to classify: across every model configuration compared in Section g, its recall is the lowest of the three classes (e.g. 0.3836 for the baseline M0), which is why the class-imbalance improvement (Section f, f.2) targets this class.

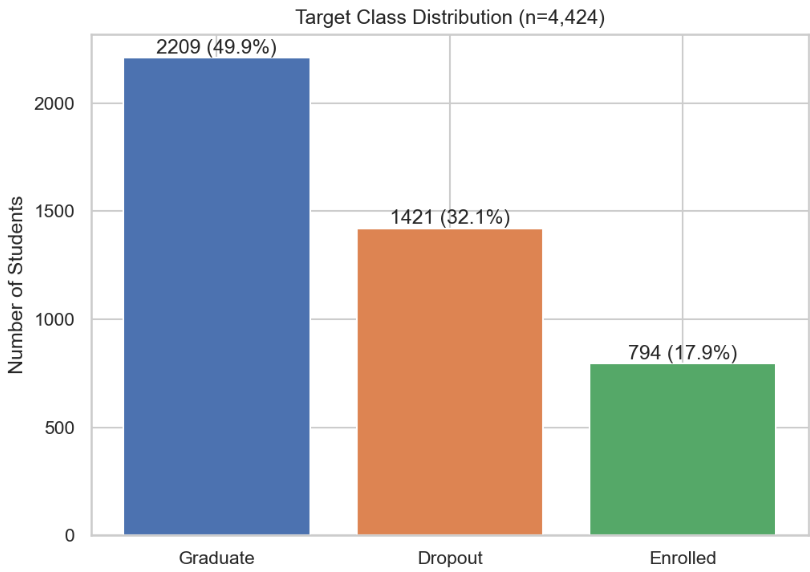

Figure 1: Target class distribution (n = 4,424).

Several of the observations below refer to how strongly a feature drives the baseline decision tree's actual splits, measured by Gini importance (the total reduction in class impurity attributed to that feature across the whole tree, described further in Section d). Table 3 lists the top 10 features by this measure, plus one lower-ranked feature included for contrast with the discussion below.

Table 3: Top-10 features by Gini importance in the baseline tree (M0), plus one feature shown for contrast

Rank	Feature	Gini importance
1	Curricular units 2nd sem (approved)	0.340
2	Curricular units 2nd sem (grade)	0.045
3	Tuition fees up to date	0.045
4	Curricular units 2nd sem (enrolled)	0.045
5	Admission grade	0.044
6	Course	0.044
7	Curricular units 1st sem (evaluations)	0.034
8	Previous qualification (grade)	0.031
9	Curricular units 1st sem (approved)	0.029
10	Mother's occupation	0.028
	...	
24	Scholarship holder	0.010

Descriptive statistics from exploratory data analysis illustrate how strongly some features relate to the target: among students who had **not paid tuition on time**, 86% ended up in the Dropout class, versus 25% Dropout / 56% Graduate among students who had paid on time. Students with a **scholarship** have a Dropout rate 26.5 percentage points lower than those without (12.2% vs. 38.7%); their rate is about one-third as high.

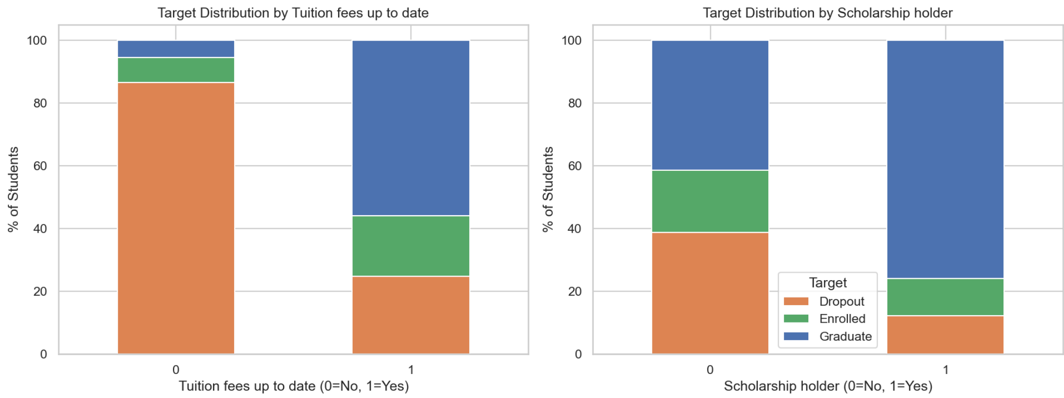

Figure 2: Target distribution split by `Tuition fees up to date` (left) and `Scholarship holder` (right).

`Tuition fees up to date` is among the top-5 features by permutation importance in the original paper (Realinho et al., 2022), and is also rank 3 in Table 3: the two ways of measuring importance agree here. `Scholarship holder` is a more subtle case: it is **not** in the top-5 of either the paper or Table 3 (rank 24), even though its percentage-of-target chart above looks just as striking as tuition's. A large gap in a simple percentage-of-target chart does not automatically mean a feature drives an important tree split once it sits alongside stronger, possibly overlapping features such as `Tuition fees up to date`; the tree may instead be extracting the same signal from that stronger correlate.

By program of study, the two lowest-graduation programs are course code 33 (Biofuel Production Technologies, $n = 12$: 8 Dropout, 3 Enrolled, and 1 Graduate; 66.7% Dropout) and course code 9119 (Informatics Engineering, $n = 170$, 54.1% Dropout). Only **8.3% and 8.2%** of their students graduate respectively, versus the dataset average of 49.9%. Course 33's rate is based on a very small sample ($n = 12$) and is not evidence of a stable program-level effect. The two highest-graduation programs are course code 9238 (Social Service, $n = 355$, 69.9% Graduate) and course code 9500 (Nursing, $n = 766$, 71.5% Graduate), with the lowest Dropout rates (18.3% and 15.4%).

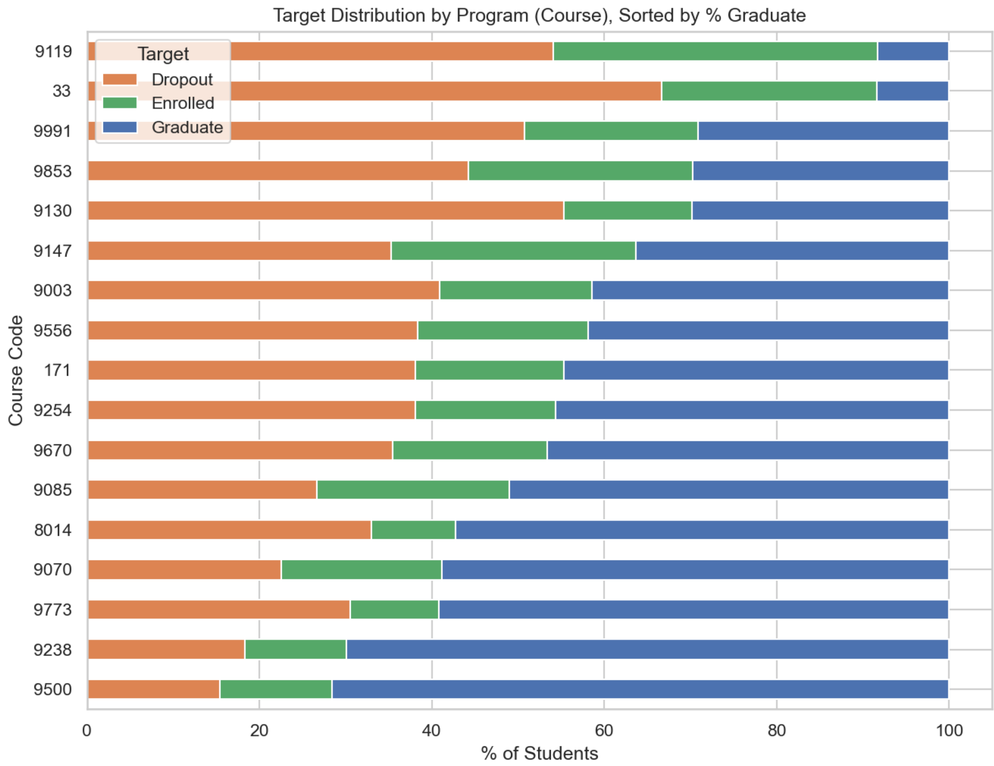

Figure 3: 17 programs sorted by increasing Graduate rate.

The original paper ranks `Course` in its top-5 features by permutation importance (Realinho et al., 2022); in Table 3, `Course` ranks 6th, close to top-5 either way. Under the study's feature-availability assumption, it is treated as available at enrollment and therefore retained in M3 (Section f, f.3).

The strongest single predictor in the whole dataset, by contrast, is only known once a semester has actually happened:

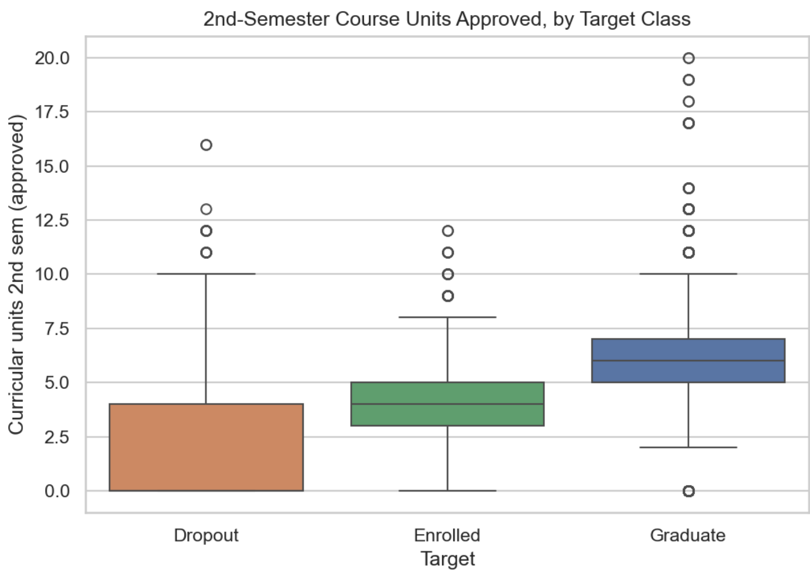

Figure 4: `Curricular units 2nd sem (approved)` by Target class.

Median values are 0 for Dropout, 4 for Enrolled, and 6 for Graduate. This is the single most important feature in Table 3 (rank 1, importance 0.340, more than the sum of ranks 2–5 combined), but it is defined as a second-semester outcome and is therefore excluded from M3.

c.3 Preprocessing performed

All preprocessing logic is implemented in a single shared pipeline, used by the whole team so every model is evaluated on the exact same train/test split.

1. **No missing-value handling needed**: the dataset is already clean at the source (0 missing values).

2. **Re-encoding categorical columns stored as integers.** Many columns are integer codes for unordered categories (e.g. `Course` code 33 = Biofuel Production Technologies, code 9500 = Nursing (Realinho et al., 2021, Variables Table, row "Course"); these codes carry no ordinal relationship). Left as-is, scikit-learn would treat them as continuous and produce meaningless splits like `Course` ≤ 8.5. The group used a mixed strategy:

   - **One-hot encode** the 4 low-cardinality columns: `Marital Status` (6), `Application mode` (18), `Course` (17), `Previous qualification` (17), so each split reads as a clear yes/no question (e.g. "is this the Nursing program?").
   - **Keep the raw integer code** for high-cardinality columns: the mother's and father's occupations (32/46 values), the mother's and father's qualifications (29/34 values), and `Nacionality` (21 values). This was a pragmatic choice to limit the number of sparse indicators. It imposes an artificial order on nominal codes, so thresholds on these columns are an acknowledged representation limitation and require cautious interpretation.
   - After this step the feature count grows from 36 to **90** columns.

3. **No feature scaling.** A decision tree splits on a threshold of one feature at a time, not on distances between data points as KNN or linear regression do, so it is invariant to any monotonic per-column transform such as scaling. Adding `StandardScaler`/`MinMaxScaler` here would be redundant and would not change the learned tree structure at all.

4. **Stratified train/test split**: 80%/20%, stratified by class to preserve the ≈50/32/18 class ratio in both sets. This guarantees that both partitions retain the observed class proportions. A fixed random seed is used so all five models in the project are evaluated on the exact same split.

After preprocessing: **3,539 training samples / 885 test samples**, each with 90 numeric features, ready to feed directly into a decision tree classifier.

c.4 Multicollinearity among features

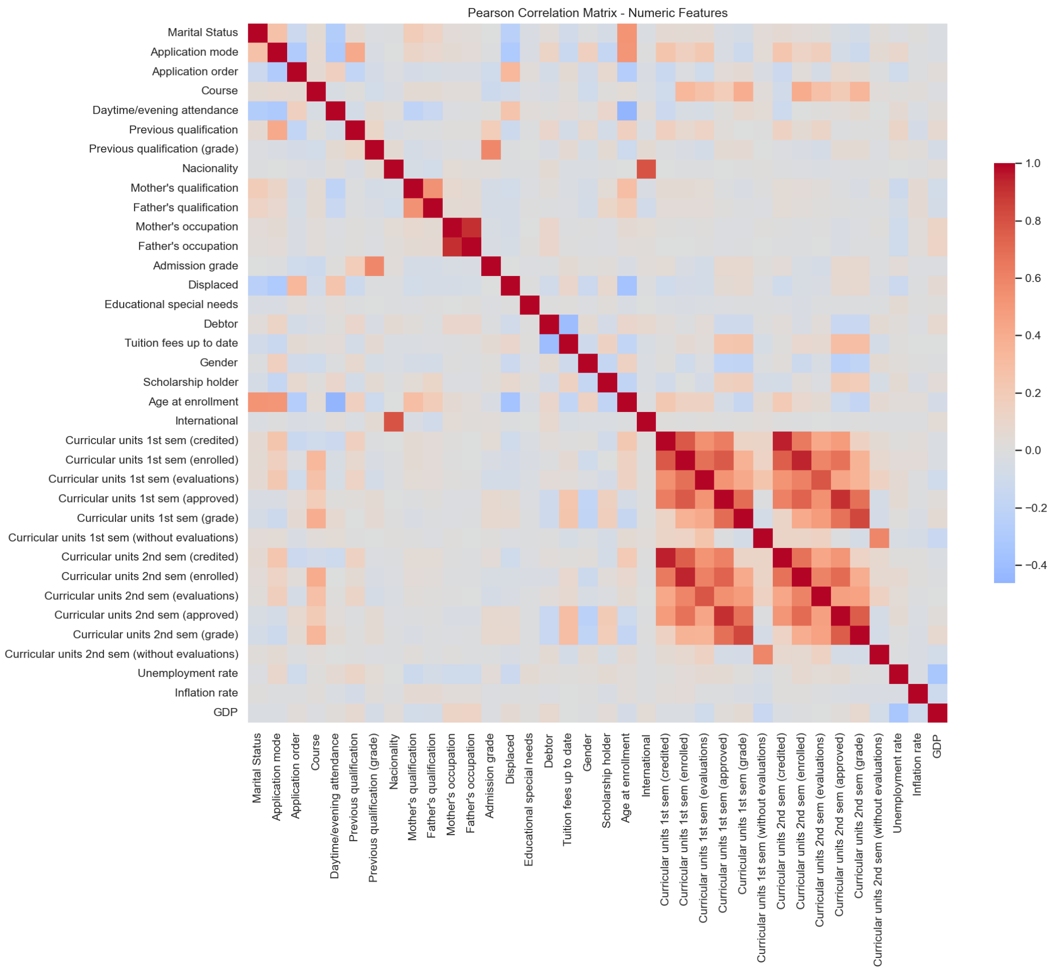

Figure 5: Pearson correlation heatmap across all numeric features.

The heatmap's colors show the overall pattern, but individual cell values are not printed on a 36×36 grid; Table 4 lists the exact values for the pairs discussed here so they can be checked directly.

Table 4: Highest Pearson correlations in the dataset

Feature	Correlated with	r
Curricular units 1st sem (credited)	Curricular units 2nd sem (credited)	0.945
Curricular units 1st sem (enrolled)	Curricular units 2nd sem (enrolled)	0.943
Mother's occupation	Father's occupation	0.910
Curricular units 1st sem (approved)	Curricular units 2nd sem (approved)	0.904
Nacionality	International	0.791

The three highest-correlation pairs in the whole dataset are `1st sem credited` ↔ `2nd sem credited`, `1st sem enrolled` ↔ `2nd sem enrolled`, and `Mother's occupation` ↔ `Father's occupation`, which is *higher* than the semester-approved pair (`1st/2nd sem approved`). The pair `Nacionality` ↔ `International` is still a high correlation (international students almost always have a non-Portuguese `Nacionality` code), but lower than the four pairs above.

A decision tree is not as sensitive to multicollinearity as linear regression (it does not need redundant features removed to converge), but multicollinearity still splits feature importance across correlated columns and can make the tree structure less stable across samples. These correlations therefore provide diagnostic context, but they are not the selection rule for M3. The canonical early-warning experiment removes exactly the 12 semester-outcome columns under the feature-availability assumption stated in Section f.3, while retaining both `Nacionality` and `International` so that feature timing is the only intentional difference from M0.

c.5 Why this dataset is appropriate for decision tree modeling

1. **Features have clear, real-world semantics** (tuition, scholarship, semester outcomes, program of study, ...), so every split translates into an ordinary sentence. This plays directly to a decision tree's interpretability strength, in a way that image or text features cannot match as easily.

2. **A mix of categorical and numerical features**: after nominal categories are encoded, the tree handles the resulting numerical and indicator columns without scaling, unlike KNN and many linear-model workflows, where feature scale directly affects distances or optimization.

3. **A genuine class imbalance** (50%/32%/18%): creates a realistic scenario for applying and comparing imbalance-handling techniques, one of the improvement directions the brief itself suggests (Section 3.2.d, "Handling class imbalance").

4. **Documented feature families support a timing experiment**: the published documentation distinguishes semester outcomes from enrollment-related data. The project therefore evaluates an early-warning exclusion rule, while treating the exact availability of every retained field as an assumption to be verified from a data dictionary and snapshot timestamps before deployment.

5. **No missing values**: lets the group spend its time on analysis and interpretation instead of data cleaning, matching the brief's own emphasis on clearly explaining how the tree is built and why the improvements help, beyond simply running a ready-made model.

c.6 Ethical note

The dataset contains sensitive variables: gender, nationality, marital status, and parents' education/occupation. Any tree splits on these variables only reflect **correlations observed in one Portuguese institution's sample from academic years 2008/09–2018/19 (Realinho et al., 2022, Section 2)**, not causal relationships, and should not be used as grounds for any discriminatory policy toward students in practice.

d. Baseline Model

d.1 Model description and training procedure

The baseline configuration, denoted M0, is an unconstrained CART classification tree implemented with scikit-learn's `DecisionTreeClassifier` (Breiman et al., 1984; Pedregosa et al., 2011). It uses the default Gini splitting criterion and a fixed random seed of 42. No maximum depth, maximum leaf count, pruning penalty, or class weighting is imposed: `max_depth=None`, `min_samples_split=2`, `min_samples_leaf=1`, and `ccp_alpha=0.0`. This deliberately high-capacity configuration provides a transparent reference against which the three improvement methods can be evaluated.

The shared preprocessing procedure described in Section c expands four lower-cardinality nominal attributes by one-hot encoding and retains the remaining numerical or encoded attributes, producing 90 model inputs. The data is divided once using a fixed, stratified 80/20 split. Table 5 shows that the class proportions are preserved across the 3,539-sample training partition and the 885-sample held-out test partition.

Table 5: Class distribution in the fixed stratified train/test split.

Class	Train count	Train share	Test count	Test share
Dropout	1,137	32.13%	284	32.09%
Enrolled	635	17.94%	159	17.97%
Graduate	1,767	49.93%	442	49.94%
Total	3,539	100.00%	885	100.00%

No feature scaling is applied. A decision tree compares values within one feature at a time, so a monotonic rescaling changes the numerical threshold but not the ordering of samples or the resulting partition. The fitted model is evaluated on the training partition to expose memorization and once on the held-out partition to estimate generalization.

d.2 Resulting tree

Figure 6 presents the complete fitted structure. Its density is useful evidence that the unconstrained model partitions the training data very finely, although individual nodes are necessarily difficult to inspect at page scale. The quantitative complexity of this tree is reported in Table 12.

Figure 6: Full structure of the unconstrained M0 decision tree.

Figure 7 limits only the visualization to the first three levels; the fitted estimator itself is unchanged. This view makes the root and early branches legible, including their split conditions, Gini impurity, sample counts, class distributions, and predicted classes. These nodes are interpreted in detail in Section e.

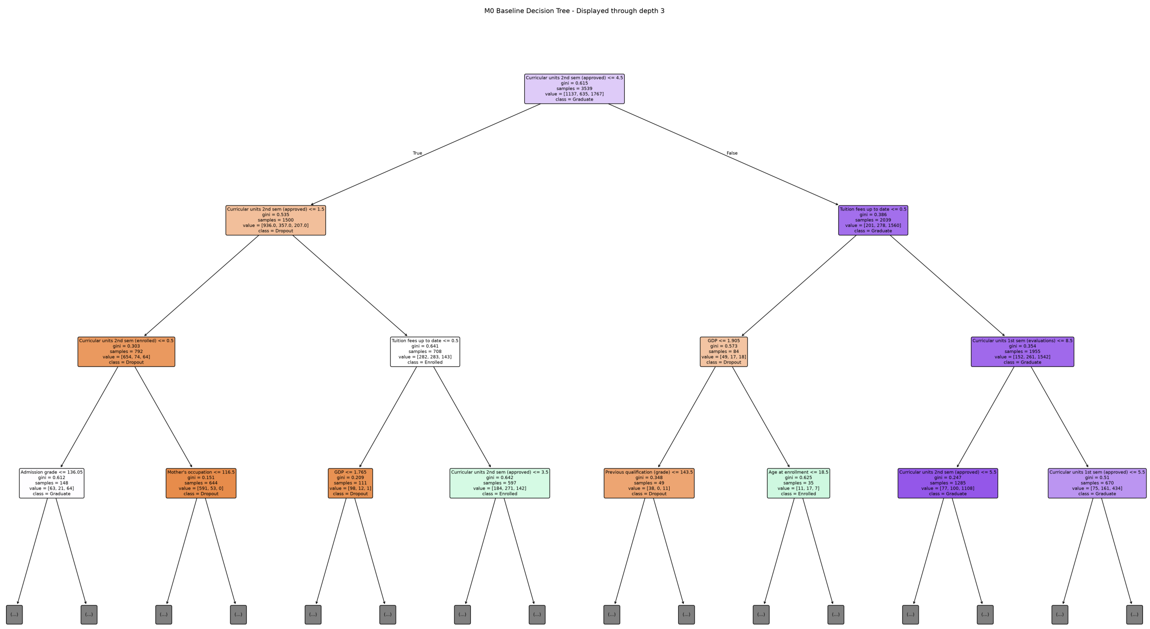

Figure 7: First three levels of the M0 decision tree.

d.3 Accuracy and error rate

Table 6 reports the complete baseline evaluation. M0 fits every training observation but correctly classifies only 66.89% of the held-out observations, yielding an error rate of 33.11%. Macro-averaged precision, recall, and F1 are all close to 0.61, below overall accuracy because each class receives equal weight and the minority Enrolled class is substantially harder to recognize. The macro one-vs-rest ROC-AUC of 0.719456 indicates moderate ranking ability across the three classes, above chance but far from strong separation.

Table 6: Performance and structural metrics of the M0 baseline.

Metric	M0
Train accuracy	1.000000
Test accuracy	0.668927
Error rate	0.331073
Macro precision	0.607642
Macro recall	0.609310
Macro-F1	0.608271
Macro ROC-AUC (OvR)	0.719456
Recall: Dropout	0.679577
Recall: Enrolled	0.383648
Recall: Graduate	0.764706
Tree depth	27
Number of leaves	634

The class-level results in Table 7 make the source of the macro-score reduction explicit. Graduate is the strongest class, with precision 0.7842 and recall 0.7647. Dropout has balanced precision and recall near 0.68. Enrolled is much weaker: only 61 of its 159 test observations are correctly identified, producing precision 0.3567, recall 0.3836, and F1 0.3697.

Table 7: Class-level performance of M0 on the held-out test partition.

Class	Precision	Recall	F1-score	Support
Dropout	0.6820	0.6796	0.6808	284
Enrolled	0.3567	0.3836	0.3697	159
Graduate	0.7842	0.7647	0.7743	442

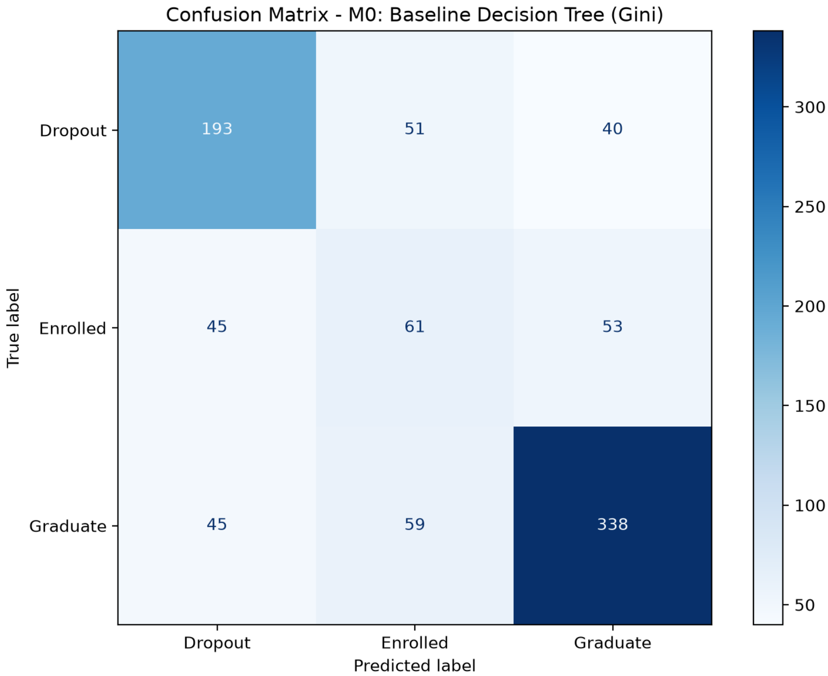

Figure 8: Confusion matrix of M0 on the 885-sample held-out test set.

Figure 8 provides the underlying counts. M0 correctly classifies 193 of 284 Dropout observations and 338 of 442 Graduate observations. Among the 159 Enrolled observations, 61 are correct, while 45 are assigned to Dropout and 53 to Graduate. Errors occur in both directions; these held-out counts describe the observed confusion but do not by themselves establish the geometry of the classes in feature space.

d.4 Gini-versus-Entropy diagnostic

Changing only the splitting criterion does not resolve the baseline's overfitting. As shown in Table 8, both configurations fit the training partition perfectly and reach depth 27. Entropy produces 50 fewer leaves, but its test accuracy is 0.015819 lower (1.58 percentage points) and its error rate is correspondingly higher. Gini is therefore retained for M0; the Entropy run is a diagnostic comparison rather than one of the three official improvement methods.

Table 8: Diagnostic comparison of Gini and Entropy on the fixed split.

Criterion	Train acc.	Test acc.	Error rate	Depth	Leaves
Gini (official M0)	1.000000	0.668927	0.331073	27	634
Entropy (diagnostic)	1.000000	0.653107	0.346893	27	584

e. Analysis of the Tree

e.1 Root split

The root split is `Curricular units 2nd sem (approved)` ≤ 4.5. Before the split, the root contains 3,539 training observations with class distribution $[1{,}137, 635, 1{,}767]$ in the order [Dropout, Enrolled, Graduate] and Gini impurity 0.615292. Table 9 shows how the split separates a left branch dominated by Dropout from a right branch dominated by Graduate.

Table 9: Sample and class distributions at the root and its two children.

Node/condition	n	[Dropout, Enrolled, Graduate]	Gini	Prediction
Root	3,539	[1,137, 635, 1,767]	0.615292	Graduate
Approved ≤ 4.5	1,500	[936, 357, 207]	0.534936	Dropout
Approved > 4.5	2,039	[201, 278, 1,560]	0.386345	Graduate

Weighting the two child impurities by their sample counts gives

$$G_{\text{after}} = \frac{1{,}500}{3{,}539}(0.534936) + \frac{2{,}039}{3{,}539}(0.386345) = 0.449325, \tag{1}$$

and therefore

$$\Delta G = 0.615292 - 0.449325 = 0.165967. \tag{2}$$

CART evaluates candidate feature-threshold pairs at a node and selects the pair that maximizes this weighted impurity reduction (Breiman et al., 1984). Thus, the direct reason for the root choice is not merely that the feature is "important": among the candidate splits, the 4.5 threshold creates the strongest reduction in class mixing. In context, students with fewer approved second-semester units are concentrated in the Dropout branch, whereas those above the threshold are concentrated in the Graduate branch. This is an association in the observed data, not evidence that crossing this threshold causally determines a student's outcome.

e.2 Root and the next three levels

Tables 10 and 11 trace every node from depth 0 through depth 3. The notation $[D, E, G]$ denotes counts for Dropout, Enrolled, and Graduate. The early structure repeatedly combines academic progress with tuition status: semester approvals establish the broad risk groups, while tuition, earlier evaluations, grades, and a small number of context variables refine those groups.

Table 10: Root through depth 2 of the M0 decision tree.

Node	Path to node	Split at node	n; $[D, E, G]$	Prediction	Interpretation
0/0	Root	`Curricular units 2nd sem (approved)` ≤ 4.5	3,539; [1,137, 635, 1,767]	Graduate	Separates lower from higher second-semester progress.
1/1	Approved ≤ 4.5	`Curricular units 2nd sem (approved)` ≤ 1.5	1,500; [936, 357, 207]	Dropout	Isolates the group with almost no approved units.
1/552	Approved > 4.5	`Tuition fees up to date` ≤ 0.5	2,039; [201, 278, 1,560]	Graduate	Tuition status refines a high-progress branch dominated by Graduate.
2/2	Approved ≤ 1.5	`Curricular units 2nd sem (enrolled)` ≤ 0.5	792; [654, 74, 64]	Dropout	Distinguishes no second-semester enrollment from enrollment with very few approvals.
2/213	$1.5 <$ approved ≤ 4.5	`Tuition fees up to date` ≤ 0.5	708; [282, 283, 143]	Enrolled	Dropout and Enrolled are nearly tied in this intermediate-progress region.
2/553	Approved > 4.5; tuition not current	`GDP` ≤ 1.905	84; [49, 17, 18]	Dropout	A small branch is divided by the macroeconomic value associated with enrollment year.
2/588	Approved > 4.5; tuition current	`Curricular units 1st sem (evaluations)` ≤ 8.5	1,955; [152, 261, 1,542]	Graduate	Graduate strongly dominates; first-semester evaluation count provides further refinement.

Table 11: All depth-3 nodes of the M0 decision tree.

Node	Path to node	Split at node	n; $[D, E, G]$	Prediction	Interpretation
3/3	Enrolled units ≤ 0.5	`Admission grade` ≤ 136.050	148; [63, 21, 64]	Graduate	This small mixed node is almost tied between Dropout and Graduate.
3/98	Enrolled units > 0.5	`Mother's occupation` ≤ 116.500	644; [591, 53, 0]	Dropout	Dropout is overwhelming; the occupation code is sensitive and only correlational.
3/214	Intermediate progress; tuition not current	`GDP` ≤ 1.765	111; [98, 12, 1]	Dropout	Non-current tuition is associated with a high Dropout share in this region.
3/235	Intermediate progress; tuition current	`Curricular units 2nd sem (approved)` ≤ 3.5	597; [184, 271, 142]	Enrolled	Enrolled is largest, but substantial class mixing remains.
3/554	`GDP` ≤ 1.905	`Previous qualification (grade)` ≤ 143.500	49; [38, 0, 11]	Dropout	Prior grade further separates a small tuition-risk subgroup.
3/571	`GDP` > 1.905	`Age at enrollment` ≤ 18.500	35; [11, 17, 7]	Enrolled	This very small node remains highly mixed despite an Enrolled plurality.
3/589	First-semester evaluations ≤ 8.5	`Curricular units 2nd sem (approved)` ≤ 5.5	1,285; [77, 100, 1,108]	Graduate	Graduate dominates; the tree refines the number of second-semester approvals.
3/996	First-semester evaluations > 8.5	`Curricular units 1st sem (approved)` ≤ 5.5	670; [75, 161, 434]	Graduate	First-semester approvals help distinguish Graduate from Enrolled.

The node paths also expose an encoding limitation. A threshold of 0.5 on a one-hot indicator means absence versus presence of that category, but a threshold on a retained high-cardinality category code, such as a parental occupation, imposes an artificial numerical ordering. Such branches are valid for the fitted representation but require more caution than genuinely ordinal academic variables.

e.3 Tree complexity

Table 12 quantifies the structure visible in Figure 6. The unconstrained tree contains 1,267 nodes and 634 leaves, reaches depth 27, and makes every training leaf pure. Of those leaves, 279 contain exactly one training observation and 399 contain no more than two. Consequently, 44.01% of leaves memorize a single observation and 62.93% represent at most two observations.

Table 12: Structural complexity of the unconstrained M0 tree.

Quantity	Value
Maximum depth	27
Total nodes	1,267
Total leaves	634
Single-sample leaves	279 (44.01%)
Leaves with at most two samples	399 (62.93%)
Pure leaves	634/634 (100%)
Average leaf depth	13.323

The average terminal node occurs at depth 13.323, so the issue is not only a few isolated deep branches. A substantial portion of the tree repeatedly partitions already small subsets until class impurity reaches zero. The full tree is therefore useful as a visual demonstration of complexity, while the depth-limited view is more appropriate for interpretation.

e.4 Overfitting evidence

Table 13 brings the generalization evidence together. The tree attains 1.000000 training accuracy but only 0.668927 test accuracy, a gap of 0.331073, or 33.11 percentage points. The test accuracy is not anomalously high; however, predictive performance alone cannot rule out data leakage. Leakage control instead relies on using the fixed held-out split and fitting the model only on the training partition. The combination of perfect training fit, a large held-out gap, depth 27, and hundreds of one- or two-sample pure leaves is direct evidence of high variance and substantial overfitting.

Table 13: Generalization and complexity indicators for M0.

Indicator	Value
Train accuracy	1.000000
Test accuracy	0.668927
Generalization gap	0.331073 (33.11 pp)
Maximum depth	27
Pure leaves	634/634
Leaves with at most two samples	399

e.5 Representative IF-THEN rules

The following rules were selected systematically rather than anecdotally. For each predicted class, the largest training leaf with purity of at least 90% was chosen, and every condition from the root to that leaf is retained. Their length and perfect training purity further illustrate how finely M0 partitions the training sample. Table 14 summarizes the terminal nodes; the full paths follow.

Table 14: Terminal statistics for the three representative M0 rules.

Predicted class	Conditions	n	[Dropout, Enrolled, Graduate]	Purity
Dropout	9	189	[189, 0, 0]	100%
Graduate	16	176	[0, 0, 176]	100%
Enrolled	16	29	[0, 29, 0]	100%

Dropout rule. IF all of the following conditions hold:

1. `Curricular units 2nd sem (approved)` ≤ 4.500;
2. `Curricular units 2nd sem (approved)` ≤ 1.500;
3. `Curricular units 2nd sem (enrolled)` > 0.500;
4. `Mother's occupation` ≤ 116.500;
5. `Curricular units 2nd sem (grade)` ≤ 13.585;
6. `Curricular units 2nd sem (evaluations)` ≤ 7.500;
7. `Marital Status_3` ≤ 0.500;
8. `Previous qualification (grade)` > 99.500;
9. `Curricular units 2nd sem (evaluations)` ≤ 4.500.

THEN predict **Dropout** ($n = 189$, distribution $[189, 0, 0]$, purity 100%). The academically meaningful part of this path is the combination of very few approved second-semester units and few evaluations, which describes low observed academic progress. Parental occupation and marital status are sensitive correlates in the fitted sample and must not be interpreted as causes of dropout.

Graduate rule. IF all of the following conditions hold:

1. `Curricular units 2nd sem (approved)` > 4.500;
2. `Tuition fees up to date` > 0.500;
3. `Curricular units 1st sem (evaluations)` ≤ 8.500;
4. `Curricular units 2nd sem (approved)` > 5.500;
5. `Curricular units 2nd sem (without evaluations)` ≤ 0.500;
6. `Course_9119` ≤ 0.500;
7. `Curricular units 1st sem (credited)` ≤ 4.500;
8. `Scholarship holder` > 0.500;
9. `Admission grade` > 98.500;
10. `Course_9773` ≤ 0.500;
11. `Curricular units 1st sem (without evaluations)` ≤ 0.500;
12. `Course_9070` ≤ 0.500;
13. `Application mode_43` ≤ 0.500;
14. `Curricular units 2nd sem (evaluations)` ≤ 10.500;
15. `Father's occupation` ≤ 8.500;
16. `Curricular units 2nd sem (grade)` > 12.264.

THEN predict **Graduate** ($n = 176$, distribution $[0, 0, 176]$, purity 100%). The clearest academic signals are more than 5.5 approved second-semester units, no second-semester units without evaluation, current tuition, and a second-semester grade above 12.264. Course, scholarship, and parental-occupation conditions are additional sample-specific correlates, not causal requirements for graduation.

Enrolled rule. IF all of the following conditions hold:

1. `Curricular units 2nd sem (approved)` ≤ 4.500;
2. `Curricular units 2nd sem (approved)` > 1.500;
3. `Tuition fees up to date` > 0.500;
4. `Curricular units 2nd sem (approved)` ≤ 3.500;
5. `Age at enrollment` ≤ 26.500;
6. `Curricular units 1st sem (approved)` ≤ 5.500;
7. `Course_9853` ≤ 0.500;
8. `Mother's occupation` > 0.500;
9. `Inflation rate` > 1.000;
10. `Curricular units 1st sem (evaluations)` ≤ 19.000;
11. `Curricular units 1st sem (grade)` > 10.100;
12. `Admission grade` ≤ 123.050;
13. `Application order` ≤ 5.500;
14. `Course_9254` ≤ 0.500;
15. `Curricular units 2nd sem (without evaluations)` ≤ 2.500;
16. `Admission grade` > 113.500.

THEN predict **Enrolled** ($n = 29$, distribution $[0, 29, 0]$, purity 100%). This path describes an intermediate state: some, but not many, second-semester units are approved; first-semester grade exceeds 10.100; and tuition is current. Its small sample and 16-condition path make it less stable than the other two representative rules. It should be read as a description of one training segment, not as a universal intervention rule.

e.6 Strengths and weaknesses of the baseline tree

M0 has several genuine strengths. Its early splits translate into ordinary domain statements, it captures nonlinear relationships and interactions without specifying them in advance, and it requires no feature scaling. The root and first few levels therefore provide an interpretable overview of how academic progress and tuition status separate the classes.

Those benefits deteriorate as the tree grows. The complete structure is too large for practical presentation, many leaves describe only one or two training observations, and the large train–test gap indicates unstable generalization. Performance is particularly weak for Enrolled. Moreover, the strongest splits rely on first- and second-semester outcomes, so M0 cannot be used at enrollment, before those outcomes exist. Whether the retained M3 variables are all available at that point remains the explicit assumption in Section f.3. These limitations directly motivate pruning, imbalance handling, and the early-warning feature set evaluated in the following sections.

> **Ethical limitation.** The data includes sensitive attributes such as gender, `Nacionality`, marital status, parents' education and occupation, and financial circumstances. Splits involving these variables reflect correlations in one Portuguese institution over the observed

period; they do not establish causal effects and must not be used as grounds for discriminatory decisions. Any real deployment would require a dedicated assessment of performance disparities across sensitive groups, group-specific error monitoring, and human oversight.

f. Improvement Methods

f.1 Improvement Method 1: Cost-Complexity Pruning

f.1.1 Method description

The unconstrained baseline tree exhibits the characteristic pattern of overfitting: it fits the training data perfectly, but its accuracy drops substantially on the held-out partition and its large structure is difficult to interpret. The first improvement therefore asks whether most of that structure can be removed without sacrificing predictive performance. M1 combines two complementary forms of regularization:

1. **Cost-complexity post-pruning** removes branches whose reduction in impurity is too small to justify their additional leaves. For a subtree T, the regularized objective is

$$R_\alpha(T) = R(T) + \alpha|\widetilde{T}|, \tag{3}$$

where $R(T)$ is the total leaf impurity and $|\widetilde{T}|$ is the number of terminal nodes. Increasing α places a larger penalty on complex trees and therefore prunes more of the weakest branches (Breiman et al., 1984; scikit-learn developers, 2026b).

2. **Pre-pruning constraints** limit tree depth and require each leaf to contain a minimum number of training samples. These constraints prevent deep branches from forming around very small, potentially high-variance subsets of the training data.

All hyperparameter selection was performed only on the training partition. The held-out test partition was not consulted while generating the pruning path, selecting α, or searching the structural grid; it was used only after the complete M1 configuration had been finalized. Both exhaustive searches used `GridSearchCV`, and class proportions were preserved across folds with `StratifiedKFold` (scikit-learn developers, 2026a,c). The complete protocol is summarized in Table 15.

Table 15: Training-only model-selection protocol for M1

Stage	Procedure and outcome
Candidate generation	The cost-complexity path contained 334 points. Removing the final one-node-tree endpoint and duplicate values left 236 unique candidate values of `ccp_alpha`.
Alpha selection	Five-fold stratified cross-validation with shuffling and random seed 42; mean validation accuracy was the prespecified selection metric and macro-F1 was retained as a diagnostic metric.
Alpha tie-break	Two candidates tied on mean validation accuracy within a tolerance of 10^{-12}. The larger value was selected to prefer the simpler tree: $\alpha = 0.00148746$, with mean CV accuracy 0.747669 and mean CV macro-F1 0.672329.
Structural search	With α fixed, all 16 pairs formed by depth limits 5, 8, 10, and 15 and minimum leaf sizes 1, 5, 10, and 20 were evaluated by the same five-fold protocol.
Held-out evaluation	The selected model was refitted on the complete training partition, then evaluated on the untouched test partition. No test-set metric was used to select a hyperparameter.

f.1.2 Modified tree / model setting

Table 16 reports mean validation accuracy for every structural configuration. The values are cross-validation scores computed from training folds, not held-out test scores. Several configurations perform similarly; the selected configuration should therefore be interpreted as the best-performing candidate in this prespecified grid, not as a universally optimal setting.

Table 16: Mean cross-validation accuracy for the structural grid

	min_samples_leaf			
max_depth	1	5	10	20
5	0.741167	0.740885	0.740603	**0.748235**
8	0.747103	0.747386	0.747104	0.745689
10	0.747951	0.747669	0.747387	0.746536
15	0.747669	0.747669	0.747387	0.746536

The structural grid increases mean CV accuracy from 0.747669 for the alpha-only model to 0.748235, but mean CV macro-F1 decreases from 0.672329 to 0.654465. This divergence follows directly from optimizing overall accuracy rather than a class-balanced metric and foreshadows the Enrolled-recall trade-off observed on the held-out partition. The resulting M1 setting and its cross-validation scores are collected in Table 17. The fold-to-fold standard deviations caution against treating the small score differences in Table 16 as evidence that nearby configurations are meaningfully inferior.

Table 17: Selected hyperparameters and fitted complexity of M1

Quantity	Selected or observed value
ccp_alpha	0.00148746
max_depth	5
min_samples_leaf	20
Random seed	42
Mean CV accuracy	0.748235 ± 0.015124
Mean CV macro-F1	0.654465 ± 0.031242
Fitted tree depth	5
Fitted number of leaves	17

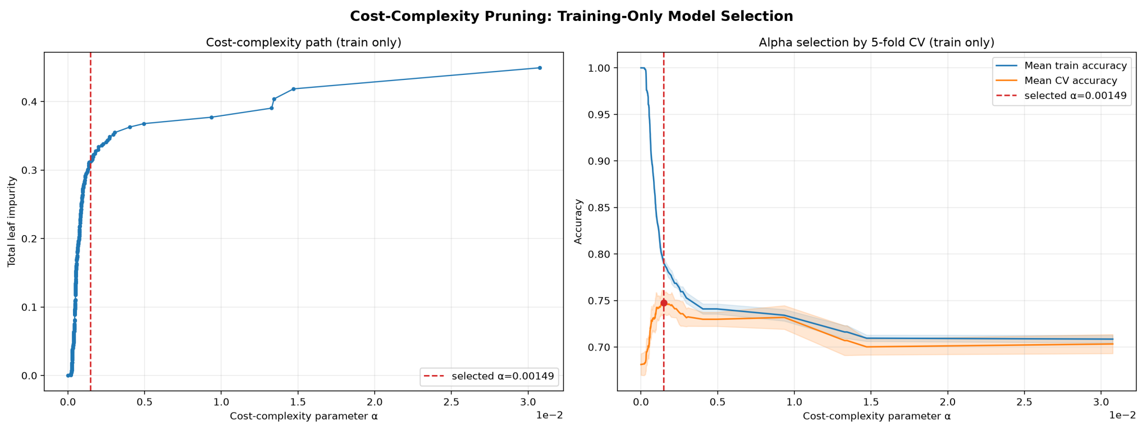

Figure 9: Cost-complexity path and cross-validation accuracy computed on the training partition.

In Figure 9, the left panel shows that total leaf impurity rises as the effective α removes progressively more structure. The right panel shows the corresponding fall in training accuracy and the validation-accuracy peak used to select α. The shaded bands show the mean score plus or minus one standard deviation across folds. No test curve is shown because the test partition was deliberately excluded from model selection.

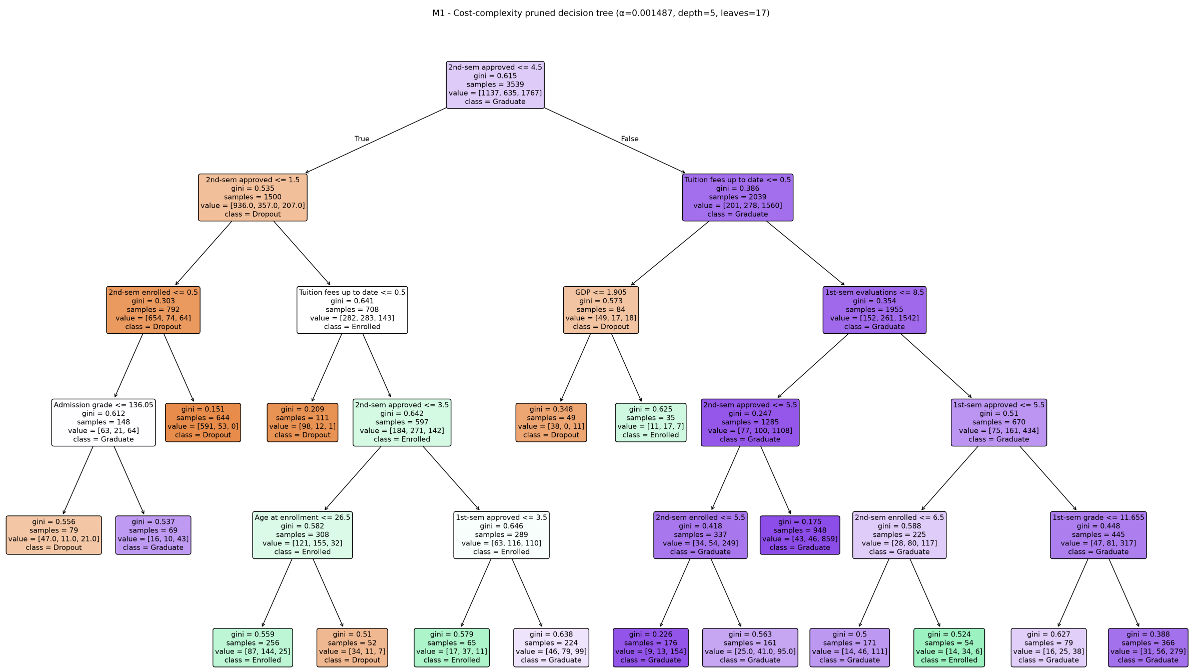

Figure 10: Complete fitted M1 tree (33 nodes, depth 5, and 17 leaves), shown without depth truncation or cropping.

Unlike the unconstrained baseline, Figure 10 presents every node and branch of the complete fitted M1 tree on a portrait page. The original source image is retained in full; individual node text may require zooming because the tree is naturally much wider than the page. Its fitted depth and leaf count are reported in Table 17.

f.1.3 Accuracy and error rate

Table 18 compares M1 with the baseline on the same held-out partition. Because error rate is defined as one minus accuracy, the 8.70-percentage-point increase in test accuracy produces an equal decrease in error rate, from 0.331073 to 0.244068. M1 also improves every aggregate metric reported here, including macro precision, macro recall, macro-F1, and macro ROC-AUC.

Table 18: Baseline and pruned-model results on the shared held-out partition

Metric	M0 baseline	M1 pruning	M1 − M0
Train accuracy	1.000000	0.768014	−0.231986
Test accuracy	0.668927	**0.755932**	+0.087006 (+8.70 pp)
Error rate	0.331073	**0.244068**	−0.087006 (−8.70 pp)
Generalization gap (train − test)	0.331073	**0.012081**	−0.318992
Macro precision	0.607642	**0.710494**	+0.102852
Macro recall	0.609310	**0.660165**	+0.050855
Macro-F1	0.608271	**0.672925**	+0.064654
Macro ROC-AUC (OvR)	0.719456	**0.847692**	+0.128237
Recall: Dropout	0.679577	**0.686620**	+0.007042
Recall: Enrolled	**0.383648**	0.345912	−0.037736
Recall: Graduate	0.764706	**0.947964**	+0.183258
Tree depth	27	**5**	−22 (−81.48%)
Number of leaves	634	**17**	−617 (−97.32%)

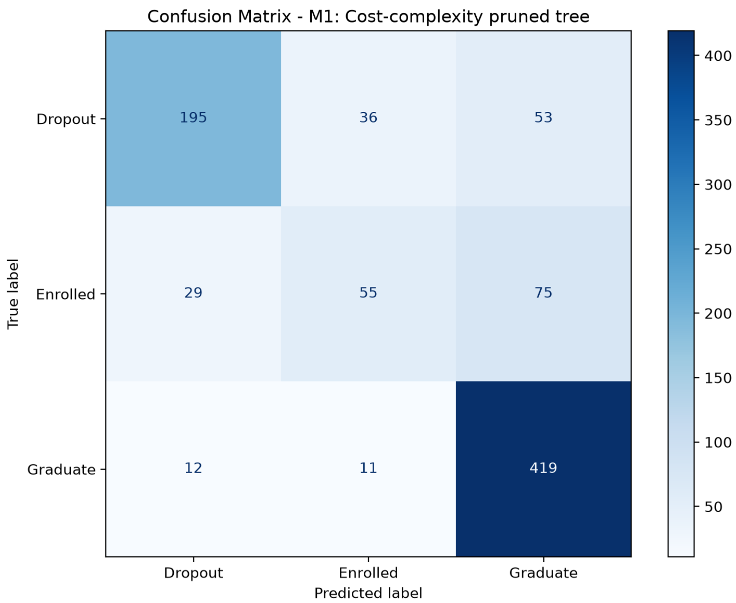

Figure 11: Confusion matrix of M1 on the held-out test partition.

Figure 11 provides the class-level counts behind the aggregate scores. M1 correctly classifies 195 of 284 Dropout students, 55 of 159 Enrolled students, and 419 of 442 Graduate students. For the Enrolled class, 29 cases are predicted as Dropout and 75 as Graduate. These counts agree with the class recalls reported in Table 18.

f.1.4 Why this improves the result

The observed pattern is consistent with the bias-variance effect expected from pruning. M0 attains perfect training accuracy by continuing to partition increasingly specific subsets of the training data. In contrast, M1 accepts higher training error in exchange for stronger regularization and a much smaller tree, which is expected to reduce variance. On the held-out split, test accuracy increases while the train–test accuracy gap decreases from 0.331073 to 0.012081; depth and leaf count also fall by 81.48% and 97.32%, respectively. Taken together, these results suggest that much of the removed structure captured sample-specific variation rather than stable predictive signal. This interpretation is supported by the current experiment but should not be generalized to every future sample without repeated evaluation.

Performance gains are not uniform across classes. Graduate recall increases from 0.764706 to 0.947964, and Dropout recall increases slightly from 0.679577 to 0.686620, whereas Enrolled recall decreases from 0.383648 to 0.345912. Because hyperparameter selection optimized overall accuracy rather than a class-balanced metric, the procedure did not explicitly protect Enrolled recall and could therefore favor the majority Graduate class. The confusion matrix supports this interpretation: 75 of the 104 misclassified Enrolled cases are assigned to Graduate. This class-specific degradation is an important trade-off and motivates the class-imbalance experiments in the next improvement method.

Finally, the observed improvement is based on one fixed held-out split and does not imply that M1 will gain exactly 8.70 percentage points on every future sample. The fold-to-fold variation in Table 17 and the close scores in Table 16 further caution against overinterpreting the exact hyperparameter choice. For this fitted experiment, however, two conclusions are directly supported: all hyperparameters were selected without consulting the held-out test partition, and M1 improved held-out aggregate performance while substantially reducing structural complexity, with a clear Enrolled-recall trade-off. Repeated evaluation would be needed to assess the stability of the exact performance gain.

f.2 Improvement Method 2: Class Imbalance

f.2.1 Motivation

The target variable of the dataset is unevenly distributed across the three outcome classes: *Graduate* is the majority class, *Dropout* is the second largest, and *Enrolled* is a clear minority. On the shared held-out test set of 885 students, the baseline model M0 correctly classifies 338 of 442 *Graduate* students and 193 of 284 *Dropout* students, but only 61 of 159 *Enrolled* students. Consequently, although M0 achieves an overall test accuracy of 0.668927, its recall on the *Enrolled* class is only 0.383648. This is a material limitation whenever the intended application requires identifying students who are not on track to graduate on time, rather than merely maximizing the number of correct predictions on the majority class.

Improvement Method 2 (M2) evaluates two complementary ways of increasing a decision tree's sensitivity to minority classes: reweighting the training objective (**M2a**) and synthetically rebalancing the training distribution (**M2b**). Both configurations reuse the project's single stratified train/test split (3,539 training rows and 885 held-out test rows, 90 features after one-hot encoding, `random_state=42`); no feature scaling and no independent re-splitting are performed in the notebook, consistent with the project-wide data contract.

f.2.2 Method description

M2a – class weighting. M2a trains a single `DecisionTreeClassifier` with `class_weight=`balanced'` and `random_state=42`, fit directly on the original (non-resampled) training data:

```
DecisionTreeClassifier(class_weight='balanced', random_state=42)
```

Under this setting, scikit-learn scales each class's contribution to the impurity criterion inversely to its frequency in the training data. No synthetic samples are created; the sample size and the feature space are identical to those used by M0.

M2b – Synthetic Minority Over-sampling Technique (SMOTE). M2b applies SMOTE (Chawla et al., 2002; imbalanced-learn developers, 2026c) to the training split only, after the shared split has already been produced, and fits an otherwise unmodified decision tree on the resampled data:

```
X_train_smote, y_train_smote = SMOTE(
sampling_strategy="auto", random_state=42, k_neighbors=5
```

```
).fit_resample(X_train, y_train)
DecisionTreeClassifier(random_state=42)
.fit(X_train_smote, y_train_smote)
```

All three SMOTE hyperparameters are declared explicitly rather than left to their library defaults, so that the resulting artifact does not silently change if a future `imbalanced-learn` release alters its default behavior (imbalanced-learn developers, 2026b). Before resampling, the training split contains 1,137 *Dropout*, 635 *Enrolled*, and 1,767 *Graduate* samples. After resampling, every class contains exactly 1,767 samples, for a total of 5,301 training rows (1,762 synthetic rows).

Leakage safeguard. SMOTE is applied strictly *after* the train/test split and is fit exclusively on `X_train`/`y_train`. Exposing the test set to resampling would cause data leakage, so the experiment also computes a value-, index-, shape-, column-order-, and dtype-aware fingerprint of `X_train`, `X_test`, `y_train`, and `y_test` immediately before calling `SMOTE.fit_resample()`, and asserts that all four fingerprints are bit-for-bit unchanged immediately afterwards. The evaluation itself is likewise performed only on the original, untouched `X_test`/ `y_test`; the resampled training data is used exclusively to fit the model, never to compute any reported metric.

f.2.3 Modified tree / model setting

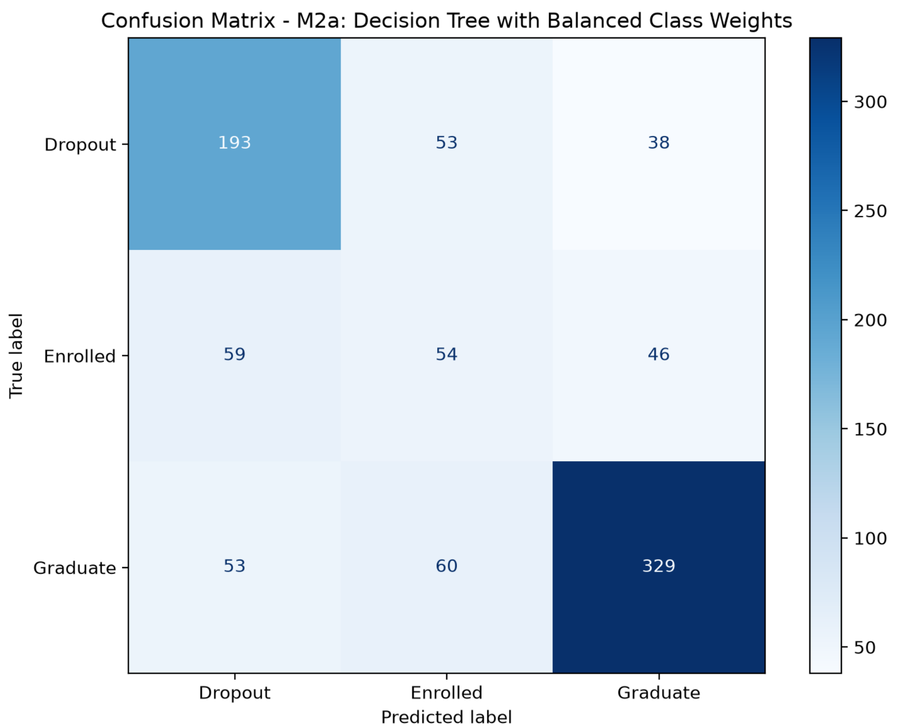

Figure 12: Confusion matrix of M2a (balanced class weights) on the 885-sample held-out test set.

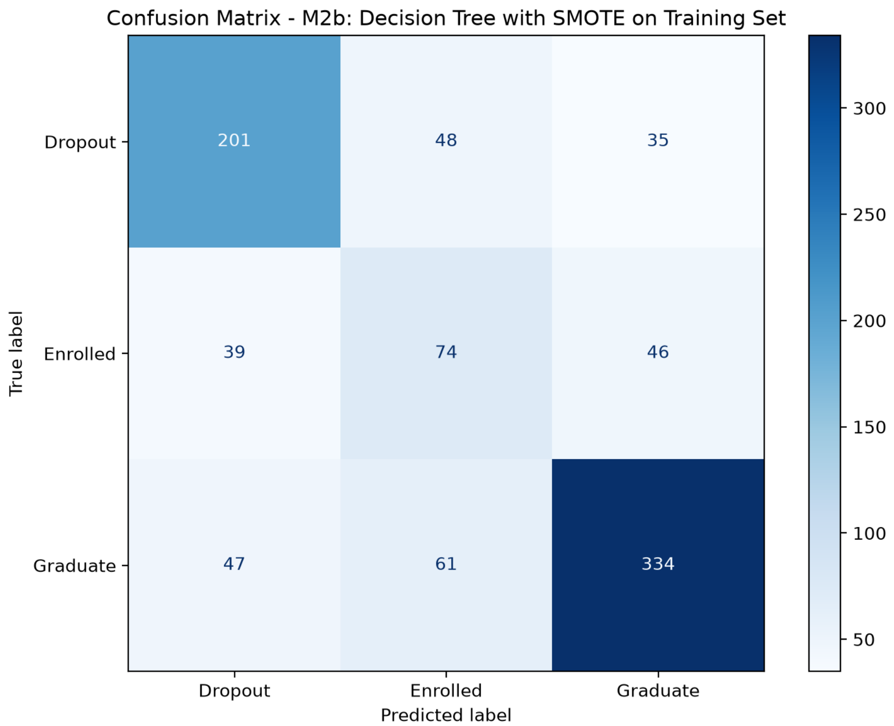

Figure 13: Confusion matrix of M2b on the 885-sample held-out test set.

M2b correctly classifies 201 Dropout and 74 Enrolled students, exceeding both M0 (193 and 61) and M2a (193 and 54). It correctly classifies 334 Graduate students, more than M2a (329) but fewer than M0 (338).

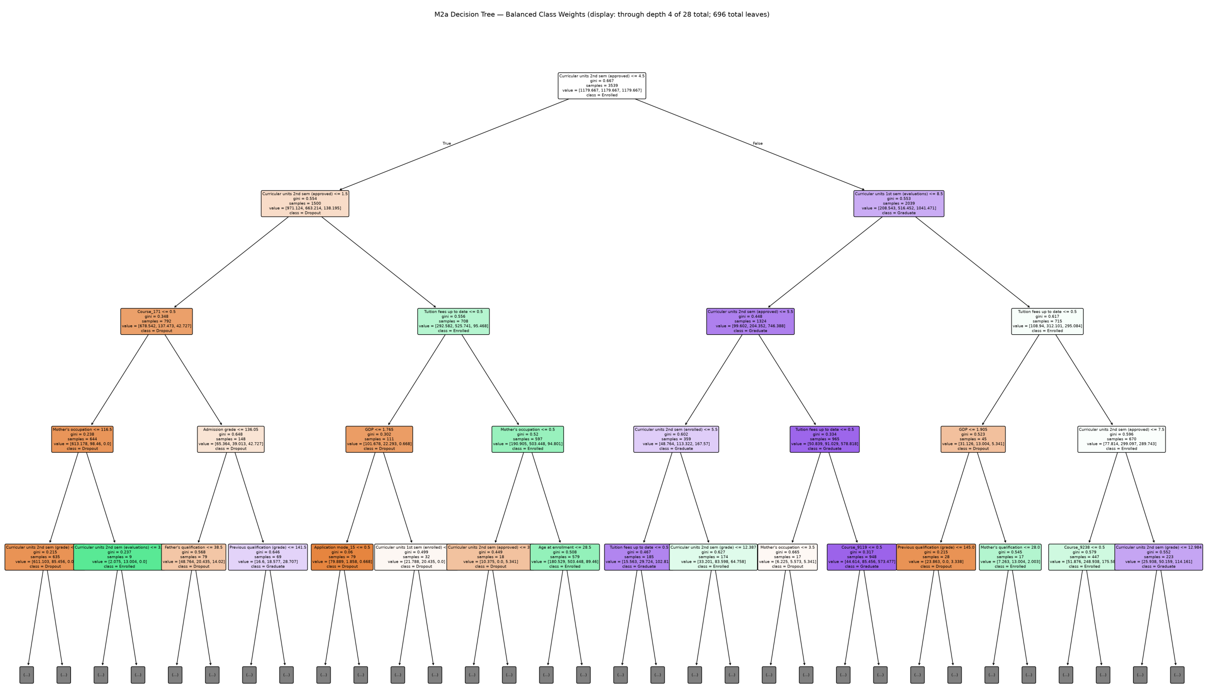

Figure 14: M2a decision tree, truncated to the first four levels for legibility. The fitted tree has a true depth of 28 and 696 leaves.

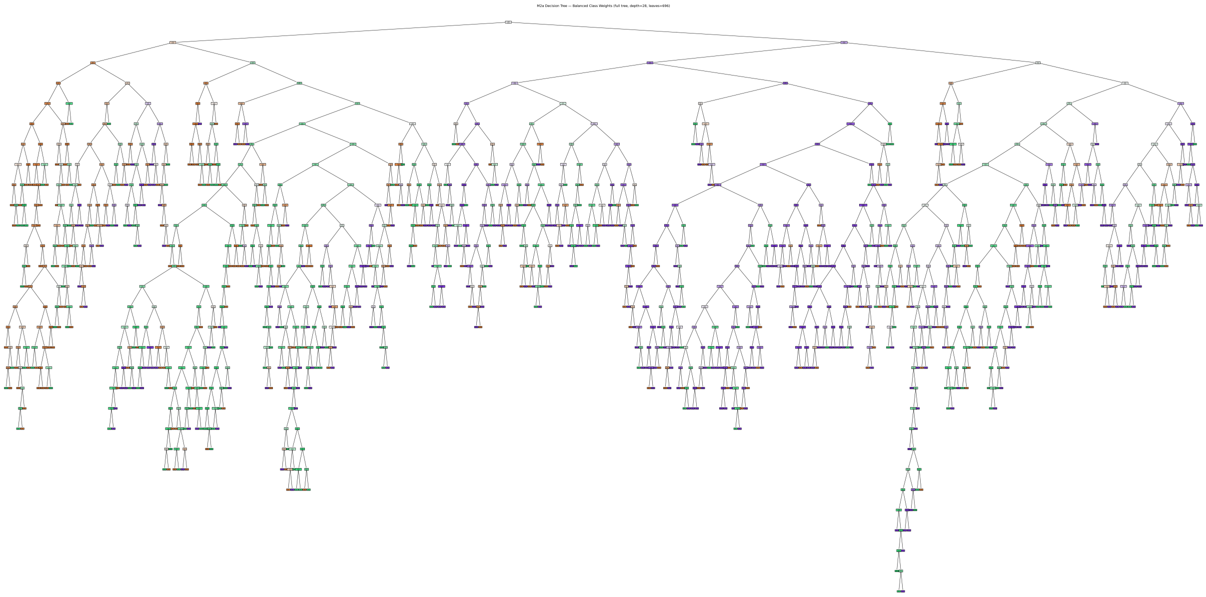

Figure 15: Full M2a decision tree (depth 28, 696 leaves).

The full M2a view is not intended for node-by-node reading. Reweighting the classes changes which splits the tree prioritizes, but the fitted tree remains close to M0 in structural size (depth 27 and 634 leaves for M0).

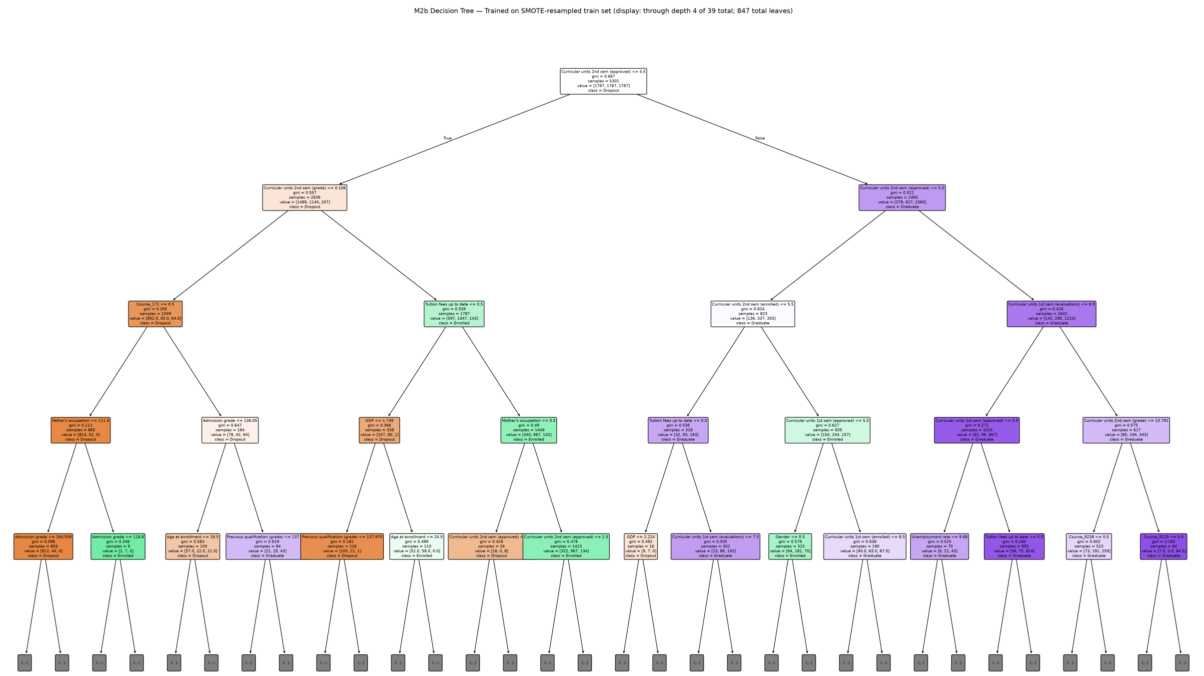

Figure 16: M2b decision tree displayed through the first four levels.

M2b was trained on the SMOTE-resampled training set of 5,301 rows and evaluated exclusively on the original held-out set. Its fitted depth is 39, with 847 leaves; the display limit does not alter the model.

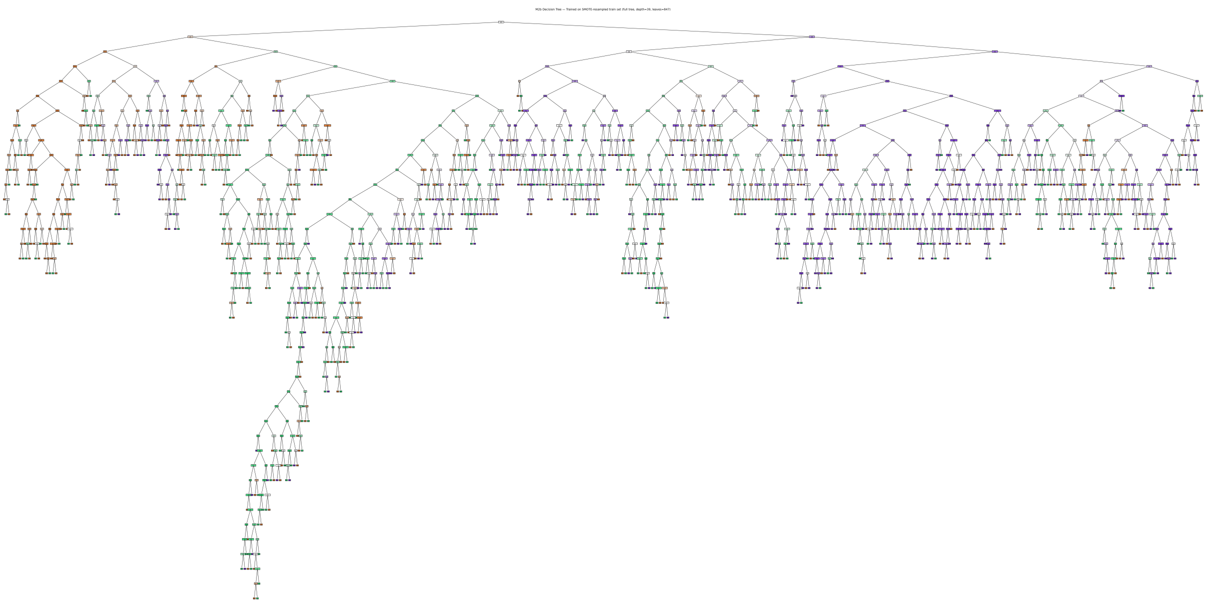

Figure 17: Full M2b decision tree (depth 39, 847 leaves).

The fitted M2b tree is larger than both M0 and M2a. SMOTE increases the training set from 3,539 to 5,301 rows, but this experiment does not isolate resampled row count as the sole cause of the increased structural complexity.

f.2.4 Accuracy and error rate

Table 19 reports the full set of held-out test metrics for M0, M2a, and M2b, together with the change relative to the baseline. Table 20 reports per-class precision and F1-score, which are not part of the shared experiment-results artifact schema but are read directly from the two classification reports.

Table 19: Held-out test performance: M0 baseline versus M2a and M2b.

Metric	M0	M2a	M2b	M2a − M0	M2b − M0
Test accuracy	0.668927	0.650847	**0.688136**	−0.018079	+0.019209
Error rate	0.331073	0.349153	**0.311864**	+0.018079	−0.019209
Recall (Dropout)	0.679577	0.679577	**0.707746**	+0.000000	+0.028169
Recall (Enrolled)	0.383648	0.339623	**0.465409**	−0.044025	+0.081761
Recall (Graduate)	**0.764706**	0.744344	0.755656	−0.020362	−0.009050
Precision (macro)	0.607642	0.584250	**0.636513**	−0.023392	+0.028871
Recall (macro)	0.609310	0.587848	**0.642937**	−0.021462	+0.033627
F1-score (macro)	0.608271	0.585409	**0.638747**	−0.022862	+0.030475
ROC-AUC (macro, OvR)	0.719456	0.705321	**0.742122**	−0.014135	+0.022667
Tree depth / leaves	27 / 634	28 / 696	**39 / 847**	–	–

Table 20: Per-class precision and F1-score for M2a and M2b.

Class	M2a Precision	M2b Precision	M2a F1	M2b F1
Dropout	0.6328	**0.7003**	0.6553	**0.7040**
Enrolled	0.3234	**0.4044**	0.3313	**0.4327**
Graduate	0.7966	0.8048	0.7696	0.7795

Notably, M2b does not merely predict *Enrolled* more often at the expense of precision: its *Enrolled* precision *also* increases relative to M2a (from 0.3234 to 0.4044), so that when M2b predicts a student as *Enrolled* it is more likely to be correct, not merely more willing to guess that label. This is the reason the *Enrolled* F1-score of M2b (0.4327) is substantially higher than that of M2a (0.3313).

f.2.5 Why this does (or does not) improve the result

M2a does not achieve the goal of this experiment on this split. Test accuracy falls by 1.81 percentage points relative to M0; recall on *Dropout* is unchanged, recall on *Enrolled* falls by 4.40 percentage points, and recall on *Graduate* falls by 2.04 percentage points. Because both overall accuracy and the recall of the classes this improvement is meant to prioritize either fall or stay flat, there is no evidence from the held-out test set that `class_weight='balanced'` alone improves M0 in this configuration. This negative result is reported in full rather than omitted, since selectively reporting only favorable configurations would misrepresent the effectiveness of class weighting for this dataset.

On this held-out split, M2b improves the target metrics. Recall on *Enrolled* rises from 0.383648 to 0.465409 (61/159 $\rightarrow$ 74/159 students correctly identified), and recall on *Dropout* rises from 0.679577 to 0.707746 (193/284 $\rightarrow$ 201/284). *Graduate* recall falls only marginally, from 0.764706 to 0.755656. All macro-averaged metrics improve simultaneously (macro F1: 0.608271 $\rightarrow$ 0.638747; macro recall: 0.609310 $\rightarrow$ 0.642937). This is consistent with the intended mechanism of SMOTE: *Enrolled* has the fewest original observations in the training data, so synthetic interpolation in its feature-space neighbourhood gives the tree more candidate split points with which to separate *Enrolled* from the other two classes. This is a mechanistic explanation specific to this experiment, not a causal claim that every dataset or split would show an identical gain. It is also worth noting explicitly that M2b's tree is substantially larger than M0's (depth 39 versus 27, 847 versus 634 leaves): the recall improvement is not the result of a simpler, more interpretable tree, since simplification is not the objective of this improvement method (that objective is addressed separately by Improvement Method 1, cost-complexity pruning).

An overall test-accuracy *increase* for M2b is a welcome but secondary finding here. Because this experiment specifically targets class-wise detection, the primary evidence is the per-class recall table above. On this held-out set, M2b has higher Dropout and Enrolled recall than both M0 and M2a.

f.2.6 Limitation: vanilla SMOTE on encoded categorical features

The shared preprocessing pipeline one-hot encodes only four categorical groups (*Marital Status*, *Application mode*, *Course*, *Previous qualification*); the remaining high-cardinality categorical variables – *Mother's/Father's qualification*, *Mother's/Father's occupation*, *Nacionality* (nominal), and *Application order* (ordinal) — are retained as integer codes. Vanilla SMOTE treats every input feature as continuous and linearly interpolates between a sample and its nearest neighbours, which can in principle generate values that do not correspond to any real category.

This effect was measured directly on the 1,762 synthetic rows that M2b actually uses for training, rather than assumed.

Table 21: Out-of-vocabulary category codes among the 1,762 synthetic rows, for the six integer-coded categorical columns.

Column	Type	Non-integer rate	Out-of-vocabulary rate
Mother's qualification	nominal	0.00%	3.01%
Father's qualification	nominal	0.00%	2.89%
Mother's occupation	nominal	0.00%	1.19%
Father's occupation	nominal	0.00%	2.04%
Nacionality	nominal	0.00%	2.38%
Application order	ordinal	0.00%	0.00%

Table 22: Interpolation outcome for the four one-hot-encoded groups, measured on the 1,762 synthetic rows. Percentages are row-wise: every synthetic row falls into exactly one of the first three columns.

One-hot group	All-zero	Multi-hot	Fractional	Any violation	Valid one-hot
Marital Status	14.7%	0.0%	0.0%	14.7%	85.3%
Application mode	65.4%	0.0%	0.0%	65.4%	34.6%
Course	**83.0%**	0.0%	0.0%	**83.0%**	17.0%
Previous qualification	24.8%	0.0%	0.0%	24.8%	75.2%

The two measurements must be read differently.

For the **six integer-coded columns**, the audit detects no non-integer values whatsoever in the synthetic data. This result must *not* be interpreted as "SMOTE does not affect these columns": it is very plausible that interpolated fractional values were truncated to integers during `fit_resample()`'s dtype-restoration step, which would make this measurement, at the level of raw values, unable to detect the interpolation that still occurred. Even where a value is an integer, it can still fail to correspond to any real category — and the out-of-vocabulary-rate column above shows exactly this: up to 3.01% of synthetic values for *Mother's qualification* are integer codes that never appear anywhere in the real training data.

For the **four one-hot groups**, the measurement is unambiguous and fully quantitative. `imbalanced-learn` 0.14.2's internal `ArraysTransformer` restores the output DataFrame to the exact dtype of its input columns via `astype()` (imbalanced-learn developers, 2026a). Because the dummy columns produced have integer dtype, any fractional value produced by SMOTE's linear interpolation between two different one-hot vectors is truncated toward zero rather than rounded. Given that a convex combination of two valid one-hot vectors can never itself exceed 1 in any single component, this truncation can only push a component down to 0, never up past 1: the audit accordingly finds a fractional-component rate of 0.0% and a multi-hot rate of 0.0% in every group, and *100% of the observed violations are all-zero vectors* (no category active) rather than any mixture of multiple categories. *Course*'s high cardinality is a plausible contributor to its high all-zero violation rate (83.0%) under this encoding-and-restoration path. The audit does not isolate category count as the sole cause.

This is a documented limitation of vanilla SMOTE when applied to mixed continuous/categorical representations; the official `imbalanced-learn` documentation recommends `SMOTENC` for exactly this situation (imbalanced-learn developers, 2026b). Vanilla SMOTE was retained as one of the group's selected class-imbalance experiments. The assignment brief suggests handling class imbalance as a possible improvement direction but does not mandate SMOTE specifically. The M2b results reported above should therefore be read together with this limitation: the recall and precision improvements are measured on the real, untouched held-out test set, but part of the synthetic training data the tree learns from does not represent a "valid" student under the true meaning of the affected categorical columns.

f.2.7 Accuracy-recall trade-off

In a class-imbalanced classification problem, overall accuracy alone cannot be the sole acceptance criterion. A model may legitimately trade a reduction in overall accuracy for less majority-class dominance; when this happens, per-class recall — not a return to M0's accuracy figure — is the correct measure of success.

M2a illustrates that this trade-off is not automatic: its accuracy falls *and* its *Enrolled* recall falls, so the result is reported plainly rather than selectively. M2b, by contrast, is not a trade-off at all on this held-out split: it simultaneously increases test accuracy (0.668927 $\rightarrow$ 0.688136), *Dropout* recall, and *Enrolled* recall, at the cost of a 0.90 percentage-point reduction in *Graduate* recall. If the evaluation priority is detecting students who are at risk of dropping out or remaining indefinitely enrolled, M2b is therefore the preferable configuration among M0, M2a, and M2b on this held-out split.

These findings are drawn from a single fixed held-out split, and the synthetic portion of M2b's training data carries the categorical-representation limitation discussed above. Before any real-world deployment, results should be validated across multiple splits or cohorts, `SMOTENC` should be considered as an additional experiment, and performance disparities across the dataset's sensitive groups should be examined. Within the scope of this experiment and the project's shared data split, however, M2b answers the improvement question affirmatively: the minority *Enrolled* class and the *Dropout* class are both recognized more reliably when SMOTE is applied strictly to the training split.

f.2.8 Conclusion

Balanced class weighting (M2a) does not improve upon the M0 baseline on this held-out test set: overall accuracy, *Enrolled* recall, and *Graduate* recall all fall, while *Dropout* recall is unchanged. Applying SMOTE to the training split only (M2b) increases *Enrolled* recall by 8.18 percentage points and *Dropout* recall by 2.82 percentage points relative to M0, while simultaneously raising *Enrolled* precision from 0.3567 to 0.4044 and lowering the overall error rate from 0.331073 to 0.311864. M2b is therefore adopted as the more successful class-imbalance configuration within the scope of this experiment. M2a is retained in this report rather than discarded, both for transparency — not every class-balancing technique improves a given model on a given dataset — and to document, alongside M2b, the quantitative categorical-representation limitation of vanilla SMOTE rather than presenting only favorable figures.

f.3 Improvement Method 3: Feature Selection for Early Warning

f.3.1 Method description

The third improvement changes the information supplied to the classifier rather than tuning its structure. Under this study's feature-availability assumption, it evaluates an enrollment-time outcome estimate intended to support early assistance. All first- and second-semester outcome variables are therefore excluded. Table 23 defines the exclusion rule; applying the six suffixes to both semester families removes exactly 12 original features.

Table 23: Semester-outcome feature families excluded from M3

Feature family	Removed measurements	Count
First-semester curricular units	credited, enrolled, evaluations, approved, grade, and without evaluations	6
Second-semester curricular units	credited, enrolled, evaluations, approved, grade, and without evaluations	6
Total		12

Under this study's feature-availability assumption, M3 treats the remaining 24 original features as available at enrollment. They include application and program information, prior academic background, tuition and scholarship status, demographics, and the three macroeconomic variables—unemployment rate, inflation rate, and GDP—associated with the year of enrollment. The `International` indicator is retained: M3 deliberately changes only semester-result inclusion, rather than combining the timing intervention with correlation-based removal.

This timing classification is an experimental assumption, not something the CSV can prove. In particular, a field such as `Tuition fees up to date` may change after enrollment. Before operational use, the institution would need an authoritative data dictionary and row-level snapshot timestamps to verify that every retained value was actually observable at the intended prediction time and was not updated later.

Feature importance and test performance were not used to choose these inputs. They are examined only after fitting, so the held-out partition does not leak into feature selection. M3 uses the same stratified train/test rows, Gini criterion, and unconstrained scikit-learn tree configuration as M0 (Pedregosa et al., 2011). Consequently, the 12-column exclusion rule is the only intentional experimental difference. Table 24 summarizes the resulting input and model contract. As for the other tree configurations, no feature scaling is applied.

Table 24: Input and model setting for the early-warning experiment

Component	M0 baseline	M3 early warning
Original features	36	24
Columns after one-hot encoding	90	78
Semester-outcome columns included	Yes	No
Split	Same stratified train/test rows	
Classifier	Unconstrained Gini decision tree	
Random seed	42	
Feature scaling	None	
Grouped permutation audit	Accuracy, 30 repeats, seed 42	

f.3.2 Modified tree and model interpretation

Figure 18 displays M3 through depth 4 for readability; the display limit does not prune or otherwise modify the fitted model. The root tests whether tuition fees are up to date. Its next level uses GDP for the branch with overdue fees and scholarship status for the branch with fees up to date. Thus, once academic outcomes are unavailable, the upper tree shifts to financial and contextual variables treated as enrollment-time information under the stated feature-availability assumption.

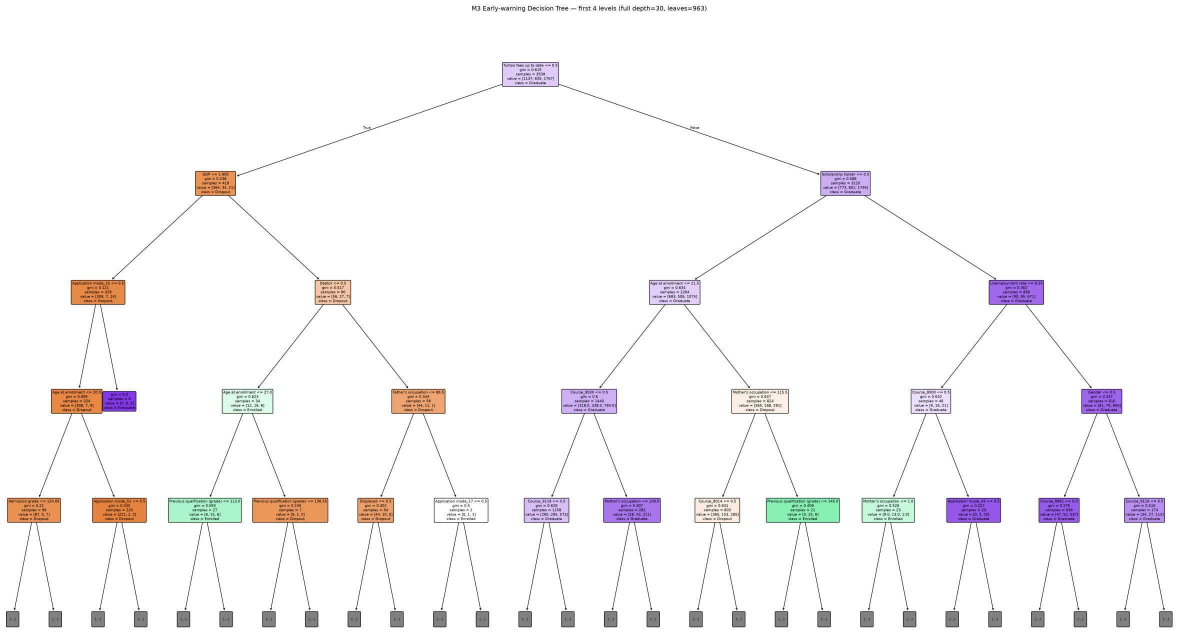

Figure 18: M3 early-warning tree displayed through depth 4.

The fitted tree has depth 30 and 963 leaves, as reported with its predictive metrics in Table 25. This is not a smaller or automatically more interpretable tree than M0: feature removal alone does not regularize an unconstrained classifier.

Figure 19 compares fitted-tree Gini/MDI importance after summing all one-hot dummy importances back to their original feature names. In M0, `Curricular units 2nd sem (approved)` dominates with an importance of 0.340. Three of M0's five highest-ranked original features are second-semester outcomes and are therefore excluded from M3. In the early-warning tree, importance is redistributed mainly toward `Course` (0.118), `Admission grade` (0.113), and `Tuition fees up to date` (0.112). These values describe how the fitted trees use their training inputs; they are associations, not causal effects, and MDI can favor features offering many possible splits.

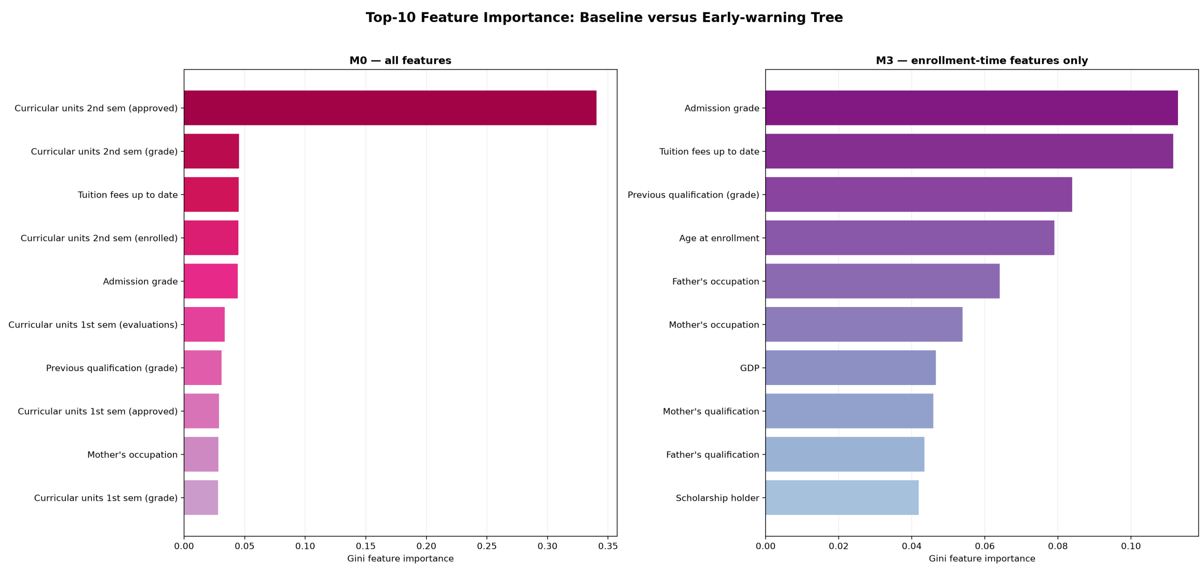

Figure 19: Top-10 aggregated Gini/MDI importances for M0 and M3.

To check whether the fitted-tree ranking also matters on unseen rows, grouped permutation importance was computed on the untouched held-out partition. All dummy columns belonging to one original categorical feature were permuted together, preserving valid one-hot groups. Figure 20 shows the mean held-out accuracy decrease and its standard deviation across 30 repetitions. The baseline remains most sensitive to second-semester approvals, whereas M3 is most sensitive to tuition status, course, and age at enrollment. The agreement between the training-derived and held-out views strengthens the interpretation without turning either importance measure into evidence of causality.

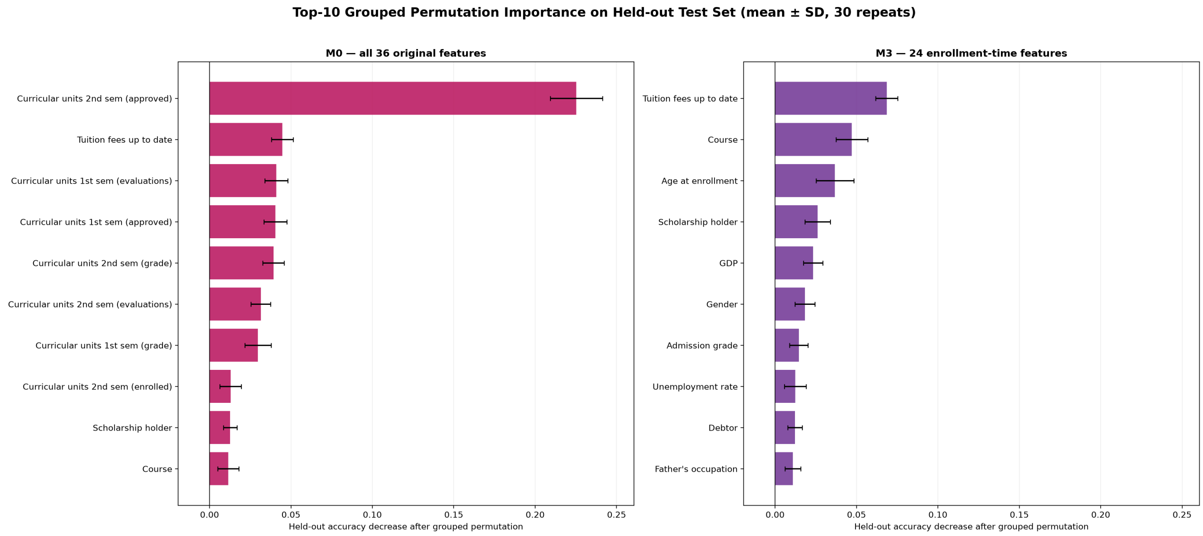

Figure 20: Grouped permutation importance on the held-out partition.

f.3.3 Accuracy and error rate

Table 25 compares M3 with M0 on the same held-out rows. M3 achieves test accuracy 0.541243 and error rate 0.458757, compared with 0.668927 and 0.331073 for M0. The 0.127684 decrease in accuracy is therefore an equal 12.77-percentage-point increase in error rate. Macro-F1 also falls from 0.608271 to 0.493050, showing that the loss is not confined to the majority class.

Table 25: Baseline and early-warning results on the shared held-out partition

Metric	M0 baseline	M3 early warning	M3 − M0
Train accuracy	1.000000	1.000000	0.000000
Test accuracy	**0.668927**	0.541243	−0.127684 (−12.77 pp)
Error rate	**0.331073**	0.458757	+0.127684 (+12.77 pp)
Generalization gap (train − test)	**0.331073**	0.458757	+0.127684
Macro precision	**0.607642**	0.492090	−0.115552
Macro recall	**0.609310**	0.494390	−0.114921
Macro-F1	**0.608271**	0.493050	−0.115221
Macro ROC-AUC (OvR)	**0.719456**	0.625355	−0.094101
Recall: Dropout	**0.679577**	0.531690	−0.147887
Recall: Enrolled	**0.383648**	0.327044	−0.056604
Recall: Graduate	**0.764706**	0.624434	−0.140271
Tree depth	27	30	+3
Number of leaves	634	963	+329

Figure 21 and Table 26 provide the class-level evidence behind the aggregate scores. M3 correctly identifies 151 of 284 Dropout students, 52 of 159 Enrolled students, and 276 of 442

Graduate students. Enrolled remains the hardest class: more of its cases are assigned to Graduate (60) or Dropout (47) than are correctly classified (52).

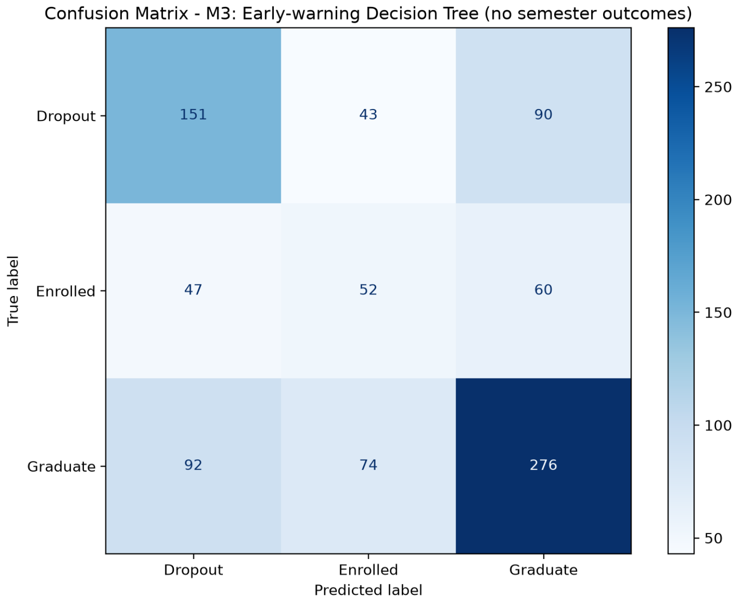

Figure 21: M3 confusion matrix on the shared held-out partition.

Table 26: Class-level performance of M3 on the held-out partition

Class	Precision	Recall	F1-score	Support
Dropout	0.5207	0.5317	0.5261	284
Enrolled	0.3077	0.3270	0.3171	159
Graduate	0.6479	0.6244	0.6359	442
Macro average	0.4921	0.4944	0.4931	885

f.3.4 Why accuracy does not improve — and why this is expected

M3 does not improve predictive accuracy because M0 has a substantial information advantage. Several of M0's strongest predictors are semester outcomes, especially second-semester approvals. They are powerful precisely because they summarize academic progress close to the eventual outcome, but they are defined after enrollment and cannot support a prediction at that earliest point. The held-out permutation analysis reaches the same qualitative conclusion: disrupting second-semester approvals causes the largest accuracy loss for M0.

The experiment also preserves the unconstrained M0 tree setting to isolate the effect of timing. M3 consequently fits the training data perfectly while generalizing poorly, with a 0.458757 train–test accuracy gap. Its greater depth and leaf count show that removing variables is not a substitute for pruning: when the strongest signals disappear, the tree can fragment the training data using weaker associations among the retained variables.

The result should therefore be interpreted as a performance–timeliness trade-off, not as a failed experiment. M0 is the stronger classifier once semester outcomes exist. Under this study's feature-availability assumption, M3 is the only configuration in this comparison designed to operate at enrollment; actual operational feasibility still requires the data-dictionary and snapshot-timestamp verification described above. This distinction is central to early student performance prediction, where the value of a warning depends partly on whether it arrives early enough for an intervention to be possible (Martins et al., 2021).

Finally, M3 is an interpretable laboratory experiment rather than a production-ready decision system. Its demographic and socioeconomic splits reflect associations in the observed sample, not causal effects. An early-warning output should support human review and offers of assistance; it should not be used as an automatic or punitive decision about a student.

g. Comparison of Results

g.1 Overall comparison table

Table 27: All 5 models on the shared 885-sample held-out test set

Model	Test acc	Error	F1 macro	ROC-AUC	Rec. Dropout	Rec. Enrolled	Rec. Graduate	Depth	Leaves
M0 (baseline)	0.6689	0.3311	0.6083	0.7195	0.6796	0.3836	0.7647	27	634
M1 (pruning)	**0.7559**	**0.2441**	**0.6729**	**0.8477**	0.6866	0.3459	**0.9480**	**5**	**17**
M2a (class_weight)	0.6508	0.3492	0.5854	0.7053	0.6796	0.3396	0.7443	28	696
M2b (SMOTE)	0.6881	0.3119	0.6387	0.7421	**0.7077**	**0.4654**	0.7557	39	847
M3 (early warning)	0.5412	0.4588	0.4931	0.6254	0.5317	0.3270	0.6244	30	963

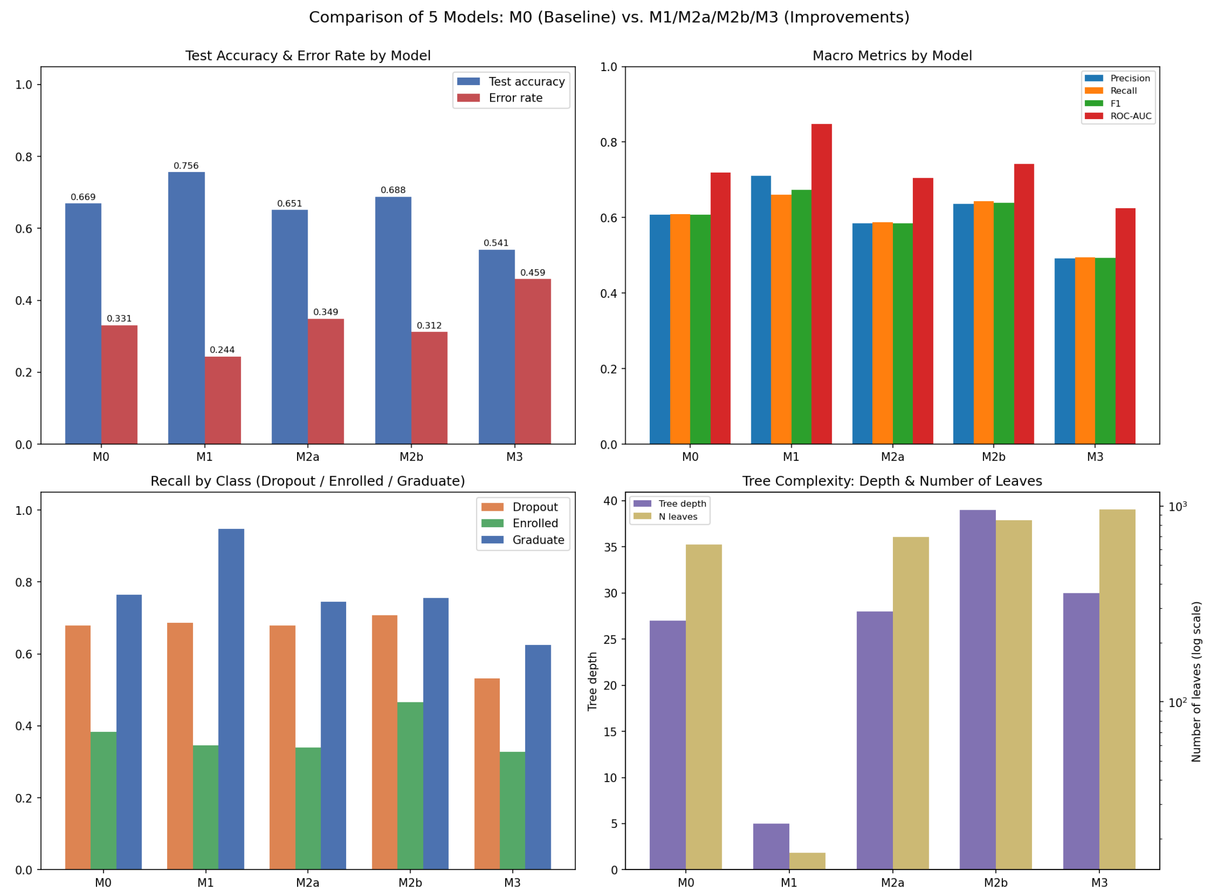

Figure 22: Four-panel comparison of M0-M1-M2a-M2b-M3 on the same held-out test set (885 samples).

Reading the four panels: top-left (test accuracy and error rate) shows M1 highest and M3 lowest; top-right (the four macro metrics: Precision, Recall, F1, ROC-AUC) shows M1 leading on all four; bottom-left (per-class recall) shows M2b dominating on Dropout and Enrolled while M1 dominates on Graduate; bottom-right (tree complexity, leaf count on a log scale) shows M1 far more compact than the other four models.

g.2 Comparing each improvement against the baseline

M1 (Pruning) improves almost every aggregate metric. Test accuracy rises from 0.6689 to 0.7559 (**+8.70 percentage points**), error rate drops from 33.11% to 24.41%, F1 macro rises 6.47 points, and ROC-AUC macro rises 12.82 points. At the same time, complexity drops sharply: depth 27→5, leaves 634→17 (**about 37× fewer**). This is a favorable result: aggregate performance and structural simplicity both improve, although Enrolled recall decreases slightly (0.3836→0.3459, −3.77 points). Accuracy was the prespecified CV criterion for selecting `ccp_alpha/max_depth/ min_samples_leaf`; because that criterion does not explicitly protect minority-class recall, it can favor the majority Graduate class whenever doing so raises the total number of correct predictions.

M2 (Class Imbalance) gives two opposite results, consistent with the experimental nature of this improvement. M2a (`class_weight='balanced'`) **does not improve** M0 on this test set: test accuracy drops 1.81 points, and even Enrolled recall, the metric this technique targets, also drops (0.3836→0.3396). In contrast, M2b (SMOTE on train only) improves both overall accuracy (+1.92 points) and the recall of the two hardest classes: Dropout recall +2.82 points, Enrolled recall **+8.18 percentage points** (61/159 → 74/159 Enrolled students correctly identified), at the cost of a slight 0.90-point drop in Graduate recall. M2b also raises Enrolled precision relative to M0 (0.3567→0.4044), so the higher recall does not simply come from the model predicting Enrolled more often. Vanilla SMOTE on encoded categorical features still has the quantitative limitations documented in Section f.2.6, so this result is best read as an improvement on this particular held-out set rather than a guarantee for a new cohort or institution.

M3 (Early Warning) drops performance substantially, and this is the expected result, not a failure. Test accuracy falls from 0.6689 to 0.5412 (−12.77 **percentage points**), F1 macro falls 11.52 points. A model-grounded explanation comes from M0's own tree rather than the original paper: by M0's Gini feature importance, 3 of the top 5 features are `2nd sem approved` (rank 1, importance 0.340, over a third of the whole tree's total importance), `2nd sem grade` (rank 2, 0.045), and `2nd sem enrolled` (rank 4, 0.045). All three fall inside the 12 columns dropped in M3 because they are only known *after* the second semester ends. The other two features in M0's real top-5, `Tuition fees up to date` (rank 3) and `Admission grade` (rank 5), are retained in M3 because they are treated as available at enrollment under this study's feature-availability assumption. The importance ranking is associative rather than causal, but the held-out performance gap is consistent with the loss of `2nd sem approved`, which alone accounts for more importance than the next four features combined. Under that assumption, M3 uses only variables treated as available at enrollment (program of study, admission grade, tuition status, the 3 macroeconomic variables, …); production use would require a data dictionary and snapshot timestamps to verify that timing.

> *The original paper ranks first-semester approvals second using permutation importance from four ensemble algorithms (Realinho et al., 2022). That is a different method from this project's M0 Gini importance. The feature ranks ninth in M0 (importance 0.029), so the different rankings are expected. The paragraph above uses only importance values measured from M0 itself.*

g.3 Which model is "best"?

There is no single answer: it depends on the criterion.

Table 28: Best model by criterion

Criterion	Best model	Why
Overall accuracy / F1 / ROC-AUC	**M1**	Highest on nearly every aggregate metric, and the most compact tree
Detecting Dropout students	**M2b**	Highest Dropout recall (0.7077)
Detecting Enrolled students (hardest class)	**M2b**	Highest Enrolled recall (0.4654), $\sim$1.21$\times$ M0
Interpretability / presentation	**M1**	Only 17 leaves, fully readable, vs. 634–963 for the other models
Early deployment under the stated timing assumption	**M3**	The only model that excludes all semester-1/2 outcomes; feasibility remains conditional on verifying retained-field timestamps

On accuracy alone, M1 wins. But the main lesson of comparing all five model configurations is that **"best" depends on the application goal.** M1 is best for an accurate, interpretable classification system used once semester data already exists. M2b is best if the goal is to miss as few at-risk students as possible (Dropout/Enrolled), accepting lower overall accuracy than M1 (0.6881 vs. 0.7559) in exchange for the better Dropout and Enrolled recall shown above; M2b still improves on M0's accuracy, so this trade-off is only relative to M1, not to the unimproved baseline. Under this study's feature-availability assumption, M3 is the only candidate for **intervention at enrollment** because it excludes every semester outcome. The other four configurations require semester data; M3's own feasibility remains conditional on verifying when each retained field is actually captured.

h. Conclusion

h.1 What the group learned

The project directly demonstrates the two theoretical points raised in the Introduction: a decision tree is both interpretable and prone to overfitting if its complexity is not controlled. After nominal categories are encoded, it also needs no feature scaling. The unconstrained M0 model reaches perfect training accuracy (1.000) while test accuracy is only 0.6689, a generalization gap of 0.3311 measured directly from this project's own results rather than assumed from theory (Breiman et al., 1984; Mitchell, 1997).

The three improvement directions highlight distinct lessons:

1. **For the severely overfit baseline in this project, controlling complexity is highly effective.** Pruning improves aggregate held-out performance and structural simplicity, although its effect is not uniform across classes because Enrolled recall decreases slightly.

2. **Class-imbalance techniques do not automatically help.** Effectiveness depends on the specific mechanism (M2a reweights the Gini computation without adding data, while M2b adds synthetic samples in the minority class's feature region), and must be judged by per-class recall, not overall accuracy alone: a model can be "worse" on accuracy yet "better" on the real application goal.

3. **The most accurate model is not always the most useful one.** M3 shows that this dataset's strongest predictive features (semester outcomes) only become available after enrollment and cannot support a prediction at that earliest decision point. Under

this study's feature-availability assumption, M3's lower accuracy quantifies the predictive cost of excluding those outcomes; actual enrollment-time use still depends on verifying retained-field timestamps.

h.2 Key findings from the experiments

- M0's overfitting: a 0.3311 generalization gap (train 1.000 vs. test 0.6689), depth 27, 634 leaves. The tree "memorizes" rather than generalizes.
- Pruning shrinks the leaf count by $\sim$**37**$\times$ (634$\to$17) while raising test accuracy by 8.70 percentage points, consistent with much of the removed structure capturing sample-specific variation rather than stable predictive signal.
- `class_weight='balanced'` (M2a) does not improve minority-class recall on this data. SMOTE-on-train (M2b) raises Enrolled recall by 8.18 points and Dropout recall by 2.82 points without a corresponding drop in Enrolled precision (Section g), so the gain is not pure over-prediction of Enrolled – though SMOTE on one-hot-encoded categorical features still carries known limitations.
- Dropping the 12 semester-outcome columns (M3) costs 12.77 accuracy points. Under the stated feature-availability assumption, this is the measured predictive cost of configuring an enrollment-time model rather than a later model that uses semester outcomes.
- All 5 models share one fixed, stratified train/test split (`random_state=42`). This makes the comparisons directly comparable and prevents different partitions from confounding the reported differences, but it does not eliminate sampling uncertainty.

h.3 Effectiveness of decision trees for this problem

A decision tree suits this dataset for the reasons argued in the Dataset Description section: features have clear semantics, so every decision rule is interpretable. For example, the Dropout rule extracted from the M0 tree in Section e is a chain of 9 conditions that together identify a leaf of $n = 189$ training samples at 100% purity. The single most readable condition in that chain is passing at most 1.5 second-semester course units (`2nd sem approved` $\leq$ 1.5), but that threshold alone is not sufficient by itself; it is one term in the full conjunction. The parallel Graduate leaf ($n = 176$, 100% purity) is a 16-condition chain, the two most readable conditions being passing more than 5.5 second-semester course units *and* `Tuition fees up to date` $>$ 0.5, consistent with the EDA observation in Figure 2 that paying tuition on time correlates strongly with graduating. Long condition chains like these are themselves a sign of the overfitting discussed above: a rule that needs 9–16 ANDed conditions to stay 100% pure is fitting narrow pockets of the training set rather than a simple general pattern, one of the concrete costs of leaving M0 unpruned. After preprocessing, the tree can split numerical and one-hot indicator columns without feature scaling. But the project also exposes the inherent limits of a single, unconstrained tree: it overfits easily, and no single model configuration is simultaneously the most accurate, the most balanced across classes' recall, and configured for the earliest prediction point; these three goals require three different configurations of the same algorithm. The class-wise recall comparison is not an assessment of performance disparities across sensitive attributes such as gender or nationality. For an application with multiple distinct goals, like dropout prediction, a more realistic conclusion than "one best model" is that **a decision tree is a flexible enough tool to be optimized for each goal separately** (accuracy, class-wise recall balance, or timeliness), provided the user chooses the right configuration and understands the trade-off being accepted. This is the kind of explanation the assignment brief asks for: why each choice helps or does not, not only which one scores highest.

References

Leo Breiman, Jerome H. Friedman, Richard A. Olshen, and Charles J. Stone. *Classification and Regression Trees*. Wadsworth, 1984.

Nitesh V. Chawla, Kevin W. Bowyer, Lawrence O. Hall, and W. Philip Kegelmeyer. SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16:321–357, 2002. doi: 10.1613/jair.953. URL https://jair.org/index.php/jair/article/view/10302. Accessed 2026-09-01.

imbalanced-learn developers. ArraysTransformer source, imbalanced-learn 0.14.2. imbalanced-learn source repository, 2026a. URL https://github.com/scikit-learn-contrib/imbalanced-learn/blob/0.14.2/imblearn/utils/_validation.py. Version 0.14.2 source; accessed 2026-09-01.

imbalanced-learn developers. Over-sampling. imbalanced-learn 0.14.2 documentation, 2026b. URL https://imbalanced-learn.org/stable/over_sampling.html. Accessed 2026-09-01.

imbalanced-learn developers. imblearn.over_sampling.SMOTE. imbalanced-learn 0.14.2 documentation, 2026c. URL https://imbalanced-learn.org/stable/references/generated/imblearn.over_sampling.SMOTE.html. Accessed 2026-09-01.

Mónica V. Martins, Daniel Tolledo, Jorge Machado, Luís M. T. Baptista, and Valentim Realinho. Early prediction of student's performance in higher education: A case study. In *Trends and Applications in Information Systems and Technologies (WorldCIST 2021)*, volume 1365 of *Advances in Intelligent Systems and Computing*, pages 166–175. Springer, 2021. doi: 10.1007/978-3-030-72657-7_16. URL https://link.springer.com/chapter/10.1007/978-3-030-72657-7_16. Accessed 2026-09-01.

Tom M. Mitchell. *Machine Learning*. McGraw-Hill, 1997.

Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexandre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Édouard Duchesnay. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011. URL https://jmlr.org/papers/v12/pedregosa11a.html. Primary article page; accessed 2026-09-01.

J. Ross Quinlan. Induction of decision trees. *Machine Learning*, 1(1):81–106, 1986. doi: 10.1007/BF00116251.

Valentim Realinho, Mónica Vieira Martins, Jorge Machado, and Luís Baptista. Predict students' dropout and academic success. UCI Machine Learning Repository, 2021. URL https://archive.ics.uci.edu/dataset/697/predict+students+dropout+and+academic+success. Dataset, id=697; accessed 2026-09-01.

Valentim Realinho, Jorge Machado, Luís Baptista, and Mónica V. Martins. Predicting student dropout and academic success. *Data*, 7(11):146, 2022. doi: 10.3390/data7110146. URL https://www.mdpi.com/2306-5729/7/11/146. Accessed 2026-09-01.

scikit-learn developers. GridSearchCV. scikit-learn documentation, 2026a. URL https://scikit-learn.org/1.9/modules/generated/sklearn.model_selection.GridSearchCV.html. Version 1.9.0 documentation; accessed 2026-09-01.

scikit-learn developers. Post pruning decision trees with cost complexity pruning. scikit-learn documentation, 2026b. URL https://scikit-learn.org/1.9/auto_examples/tree/plot_cost_complexity_pruning.html. Version 1.9.0 documentation; accessed 2026-09-01.

scikit-learn developers. StratifiedKFold. scikit-learn documentation, 2026c. URL https://scikit-learn.org/1.9/modules/generated/sklearn.model_selection.StratifiedKFold.html. Version 1.9.0 documentation; accessed 2026-09-01.